Typically, most people try to maintain the queue size the same as the window's size. Try to break away from this thought and think out of the box. Removing redundant elements and storing only elements that need to be considered in the queue is the key to achieving the efficient O(n) solution below. This is because each element in the list is being inserted and removed at most once. Therefore, the total number of insert + delete operations is 2n.

```
void MaxSlidingWindow(int A[], int n, int w, int B[]) {
    struct DoubleEndQueue *Q = CreateDoubleEndQueue();
    for (int i = 0; i < w; i++) {
            while (!IsEmptyQueue(Q) && A[i] >= A[QBack(Q)])
                    PopBack(Q);
            PushBack(Q, i);
    }
    for (int i = w; i < n; i++) {
            B[i-w] = A[QFront(Q)];
            while (!IsEmptyQueue(Q) && A[i] >= A[QBack(Q)])
                    PopBack(Q);
            while (!IsEmptyQueue(Q) && QFront(Q) <= i-w)
                    PopFront(Q);
            PushBack(Q, i);
    }
    B[n-w] = A[QFront(Q)];
}
```

Problem-31 A priority queue is a list of items in which each item has associated with it a priority. Items are withdrawn from a priority queue in order of their priorities starting with the highest priority item first. If the maximum priority item is required, then a heap is constructed such than priority of every node is greater than the priority of its children.

Design such a heap where the item with the middle priority is withdrawn first. If there are n items in the heap, then the number of items with the priority smaller than the middle priority is $\frac{n}{2}$ if n is odd, else $\frac{n}{2} \mp 1 \mp 1$.

Explain how withdraw and insert operations work, calculate their complexity, and how the data structure is constructed.

Solution: We can use one min heap and one max heap such that root of the min heap is larger than

the root of the max heap. The size of the min heap should be equal or one less than the size of the max heap. So the middle element is always the root of the max heap.

For the insert operation, if the new item is less than the root of max heap, then insert it into the max heap; else insert it into the min heap. After the withdraw or insert operation, if the size of heaps are not as specified above than transfer the root element of the max heap to min heap or vice-versa.

With this implementation, insert and withdraw operation will be in O(*logn*) time.

Problem-32 Given two heaps, how do you merge (union) them?

Solution: Binary heap supports various operations quickly: Find-min, insert, decrease-key. If we have two min-heaps, H1 and H2, there is no efficient way to combine them into a single min-heap.

For solving this problem efficiently, we can use mergeable heaps. Mergeable heaps support efficient union operation. It is a data structure that supports the following operations:

- Create-Heap(): creates an empty heap
- Insert(H,X,K) : insert an item x with key K into a heap H
- Find-Min(H) : return item with min key
- Delete-Min(H) : return and remove
- Union(H1, H2) : merge heaps H1 and H2

Examples of mergeable heaps are:

- Binomial Heaps
- Fibonacci Heaps

Both heaps also support:

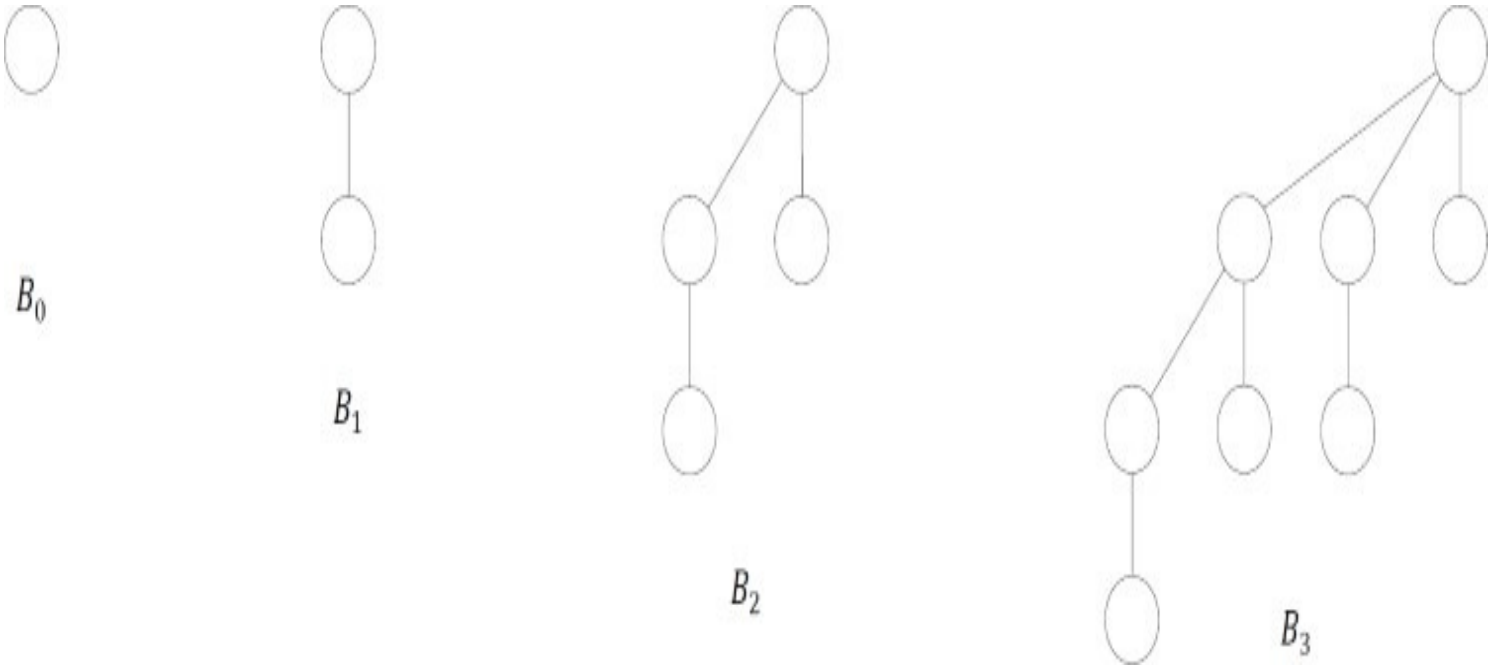
- Decrease-Key(H,X,K): assign item Y with a smaller key K
- Delete(H,X) : remove item X

Binomial Heaps: Unlike binary heap which consists of a single tree, a *binomial* heap consists of a small set of component trees and no need to rebuild everything when union is performed. Each component tree is in a special format, called a *binomial tree*.

A binomial tree of order *k*, denoted by B_k is defined recursively as follows:

- B_0 is a tree with a single node
- For $k \geq 1$, B_k is formed by joining two B_{k-1}, such that the root of one tree becomes the leftmost child of the root of the other.

Example:



Fibonacci Heaps: Fibonacci heap is another example of mergeable heap. It has no good worst-case guarantee for any operation (except Insert/Create-Heap). Fibonacci Heaps have excellent amortized cost to perform each operation. Like *binomial* heap, *fibonacci* heap consists of a set of min-heap ordered component trees. However, unlike binomial heap, it has

- No limit on number of trees (up to O(n)), and
- No limit on height of a tree (up to O(n))

Also, *Find-Min, Delete-Min, Union, Decrease-Key, Delete* all have worst-case O(n) running time. However, in the amortized sense, each operation performs very quickly.

Operation	Binary Heap	Binomial Heap	Fibonacci Heap
Create-Heap	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
Find-Min	$\Theta(1)$	$\Theta(logn)$	$\Theta(1)$
Delete-Min	$\Theta(logn)$	$\Theta(logn)$	$\Theta(logn)$
Insert	$\Theta(logn)$	$\Theta(logn)$	$\Theta(1)$
Delete	$\Theta(logn)$	$\Theta(logn)$	$\Theta(logn)$
Decrease-Key	$\Theta(logn)$	$\Theta(logn)$	$\Theta(1)$
Union	$\Theta(n)$	$\Theta(logn)$	$\Theta(1)$

Problem-33 Median in an infinite series of integers

Solution: Median is the middle number in a sorted list of numbers (if we have odd number of elements). If we have even number of elements, median is the average of two middle numbers in a sorted list of numbers.

We can solve this problem efficiently by using 2 heaps: One MaxHeap and one MinHeap.

1. MaxHeap contains the smallest half of the received integers
2. MinHeap contains the largest half of the received integers

The integers in MaxHeap are always less than or equal to the integers in MinHeap. Also, the number of elements in MaxHeap is either equal to or 1 more than the number of elements in the MinHeap.

In the stream if we get $2n$ elements (at any point of time), MaxHeap and MinHeap will both contain equal number of elements (in this case, n elements in each heap). Otherwise, if we have received $2n + 1$ elements, MaxHeap will contain $n + 1$ and MinHeap n.

Let us find the Median: If we have $2n + 1$ elements (odd), the Median of received elements will be the largest element in the MaxHeap (nothing but the root of MaxHeap). Otherwise, the Median of received elements will be the average of largest element in the MaxHeap (nothing but the root of MaxHeap) and smallest element in the MinHeap (nothing but the root of MinHeap). This can be calculated in O(1).

Inserting an element into heap can be done in O($logn$). Note that, any heap containing $n + 1$ elements might need one delete operation (and insertion to other heap) as well.

Example:

Insert 1: Insert to MaxHeap.
MaxHeap: {1}, MinHeap:{}

Insert 9: Insert to MinHeap. Since 9 is greater than 1 and MinHeap maintains the maximum elements.
MaxHeap: {1}, MinHeap:{9}

Insert 2: Insert MinHeap. Since 2 is less than all elements of MinHeap.
MaxHeap: {1,2}, MinHeap:{9}

Insert 0: Since MaxHeap already has more than half; we have to drop the max element from MaxHeap and insert it to MinHeap. So, we have to remove 2 and insert into MinHeap. With that it becomes:
MaxHeap: {1}, MinHeap:{2,9}
Now, insert 0 to MaxHeap.

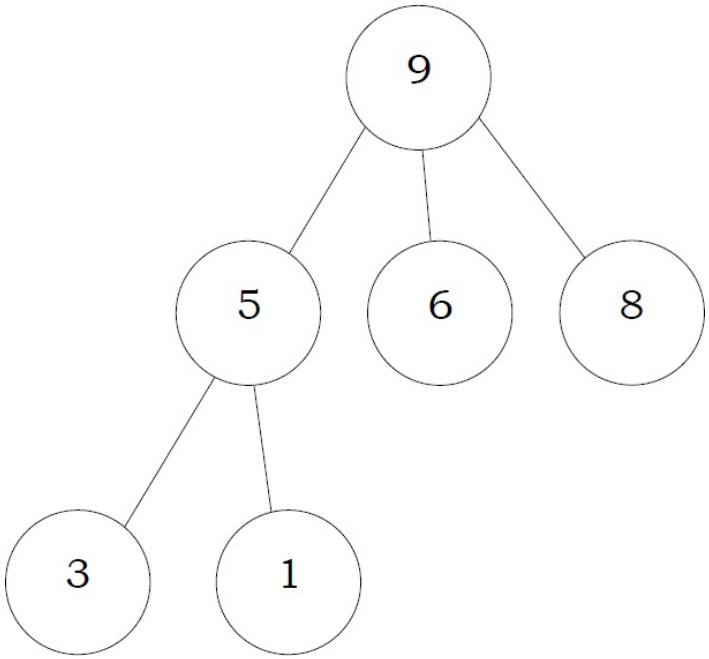
Total Time Complexity: O($logn$).

Problem-34 Suppose the elements 7, 2, 10 and 4 are inserted, in that order, into the valid 3-ary max heap found in the above question, Which one of the following is the sequence of items in the array representing the resultant heap?

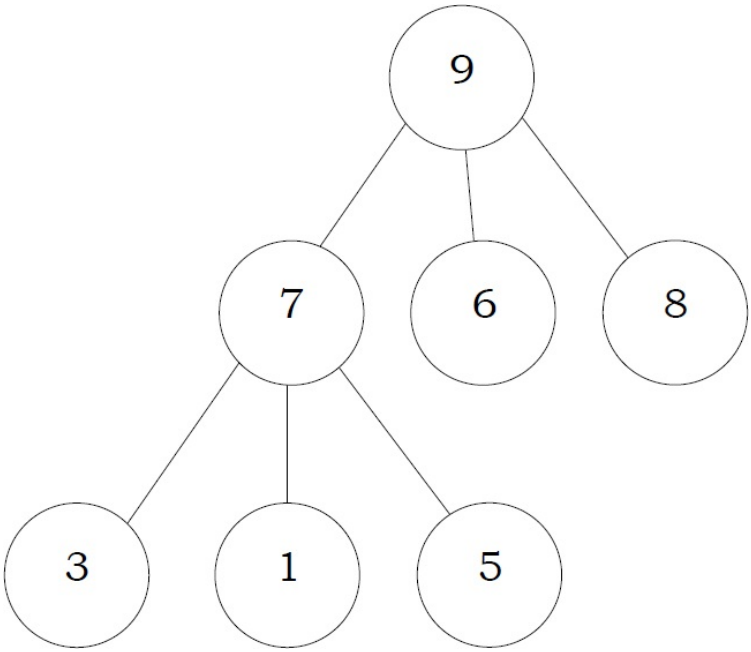
(A) 10, 7, 9, 8, 3, 1, 5, 2, 6, 4
(B) 10, 9, 8, 7, 6, 5, 4, 3, 2, 1

(C) 10, 9, 4, 5, 7, 6, 8, 2, 1, 3
(D) 10, 8, 6, 9, 7, 2, 3, 4, 1, 5

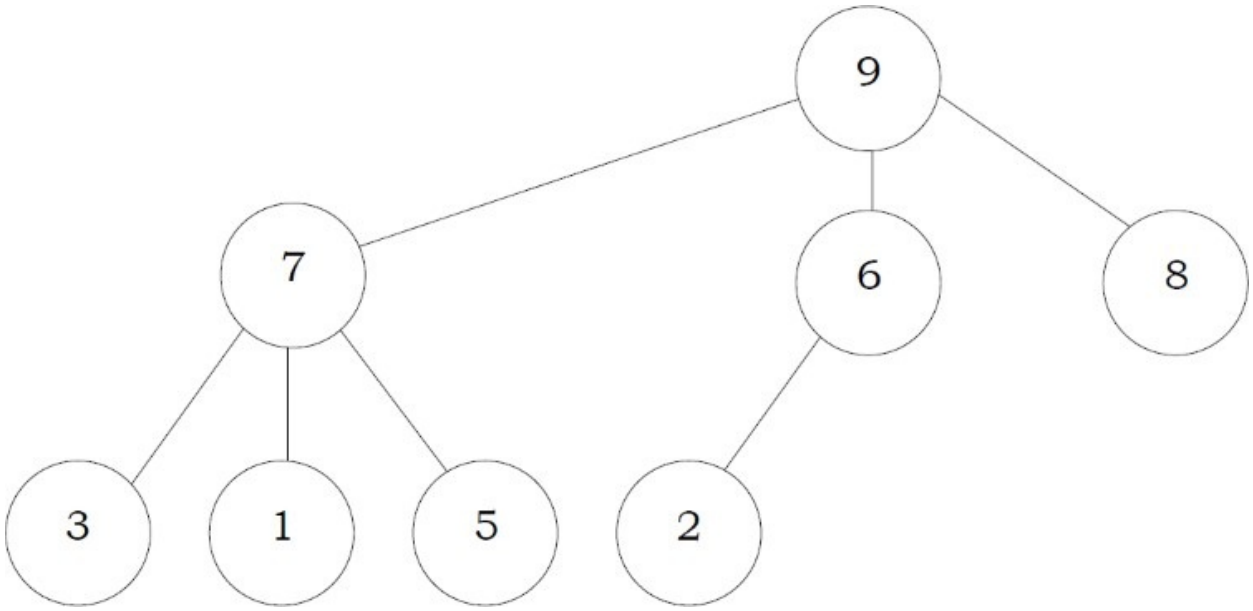
Solution: The 3-ary max heap with elements 9, 5, 6, 8, 3, 1 is:



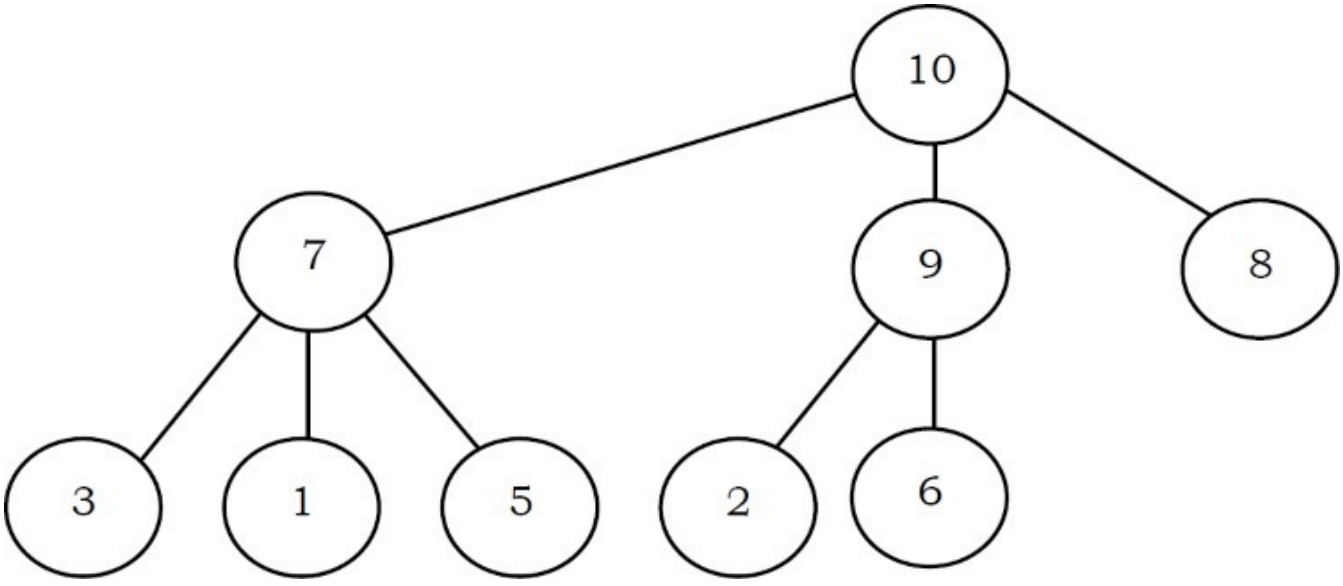
After Insertion of 7:



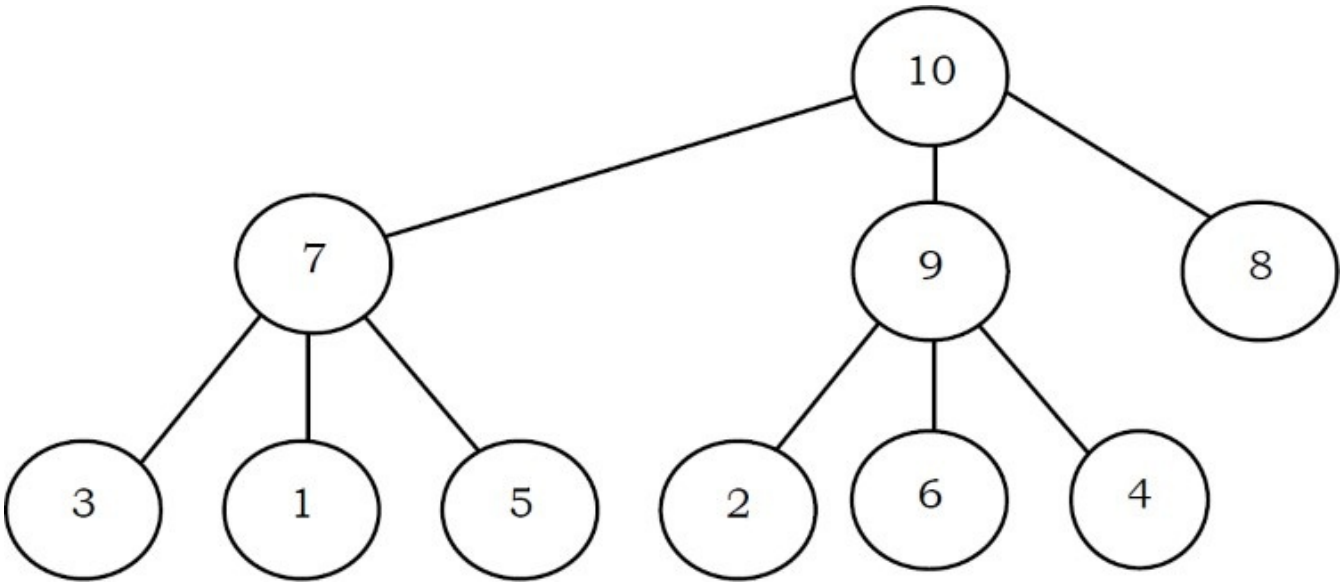
After Insertion of 2:



After Insertion of 10:

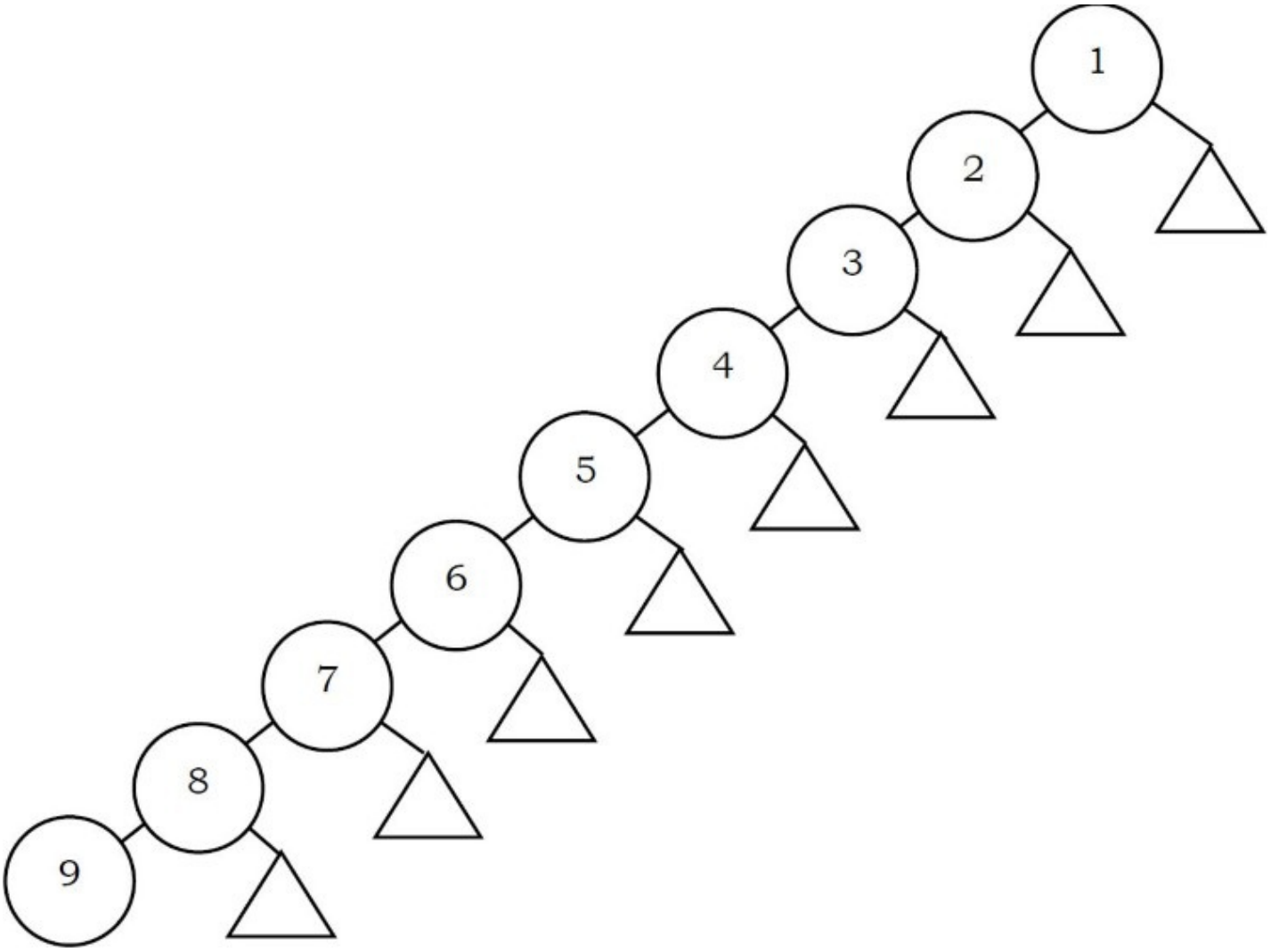


After Insertion of 4:



Problem-35 A complete binary min-heap is made by including each integer in [1,1023] exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is.

Solution: As shown in the figure below, for a given number *i*, we can fix the element *i* at i^{th} level and arrange the numbers 1 to $i - 1$ to the levels above. Since the root is at depth *zero*, the maximum depth of the i^{th} element in a min-heap is $i - 1$. Hence, the maximum depth at which integer 9 can appear is 8.



Problem-36 A *d*-ary heap is like a binary heap, but instead of 2 children, nodes have *d* children. How would you represent a *d*-ary heap with *n* elements in an array? What are the expressions for determining the parent of a given element, *Parent*(*i*), and a j^{th} child of a given element, *Child*(*i*,*j*), where 1 ≤ j ≤ d?

Solution: The following expressions determine the parent and j^{th} child of element i (where 1 ≤ j ≤ d):

$$Parent(i) = \left\lfloor \frac{i+d-2}{d} \right\rfloor$$

$$Child(i,j) = (i-1).d + j + 1$$

CHAPTER

8

DISJOINT SETS ADT

8.1 Introduction

In this chapter, we will represent an important mathematics concept: *sets*. This means how to represent a group of elements which do not need any order. The disjoint sets ADT is the one used for this purpose. It is used for solving the equivalence problem. It is very simple to implement. A simple array can be used for the implementation and each function takes only a few lines of code. Disjoint sets ADT acts as an auxiliary data structure for many other algorithms (for example, *Kruskal's* algorithm in graph theory). Before starting our discussion on disjoint sets ADT, let us look at some basic properties of sets.

8.2 Equivalence Relations and Equivalence Classes

For the discussion below let us assume that 5 is a set containing the elements and a relation R is defined on it. That means for every pair of elements in $a,b \in 5$, $a\ R\ b$ is either true or false. If $a\ R$

b is true, then we say *a* is related to *b*, otherwise *a* is not related to *b*. *A* relation *R* is called an *equivalence relation* if it satisfies the following properties:

- *Reflexive:* For every element $a \in S$.*aR a* is true.
- *Symmetric:* For any two elements $a, b \in S$, if *a R b* is true then *b R a* is true.
- *Transitive:* For any three elements a, b, c ∈ S, if a R b and *b R c* are true then *a R c* is true.

As an example, relations ≤ (less than or equal to) and ≥ (greater than or equal to) on a set of integers are not equivalence relations. They are reflexive (since $a \leq a$) and transitive ($a \leq b$ and $b \leq c$ implies $a \leq$ c) but not symmetric ($a \leq b$ does not imply $b \leq a$).

Similarly, *rail connectivity* is an equivalence relation. This relation is reflexive because any location is connected to itself. If there is connectivity from city *a* to city *b*, then city *b* also has connectivity to city *a*, so the relation is symmetric. Finally, if city *a* is connected to city *b* and city *b* is connected to city c, then city *a* is also connected to city *c*.

The *equivalence class* of an element $a \in S$ is a subset of *S* that contains all the elements that are related to *a*. Equivalence classes create a *partition* of *S*. Every member of *S* appears in exactly one equivalence class. To decide if *a R b*, we just need to check whether *a* and *b* are in the same equivalence class (group) or not.

In the above example, two cities will be in same equivalence class if they have rail connectivity. If they do not have connectivity then they will be part of different equivalence classes.

Since the intersection of any two equivalence classes is empty (ϕ), the equivalence classes are sometimes called *disjoint sets*. In the subsequent sections, we will try to see the operations that can be performed on equivalence classes. The possible operations are:

- Creating an equivalence class (making a set)
- Finding the equivalence class name (Find)
- Combining the equivalence classes (Union)

8.3 Disjoint Sets ADT

To manipulate the set elements we need basic operations defined on sets. In this chapter, we concentrate on the following set operations:

- MAKESET(*X*): Creates a new set containing a single element *X*.
- UNION(*X*, *Y*): Creates a new set containing the elements *X* and *Y* in their union and deletes the sets containing the elements *X* and *Y*.
- FIND(*X*): Returns the name of the set containing the element *X*.

8.4 Applications

Disjoint sets ADT have many applications and a few of them are:

- To represent network connectivity
- Image processing
- To find least common ancestor
- To define equivalence of finite state automata
- Kruskal's minimum spanning tree algorithm (graph theory)
- In game algorithms

8.5 Tradeoffs in Implementing Disjoint Sets ADT

Let us see the possibilities for implementing disjoint set operations. Initially, assume the input elements are a collection of n sets, each with one element. That means, initial representation assumes all relations (except reflexive relations) are false. Each set has a different element, so that $S_i \cap S_j = \phi$. This makes the sets *disjoint*.

To add the relation $a\ R\ b$ (UNION), we first need to check whether a and b are already related or not. This can be verified by performing FINDs on both a and b and checking whether they are in the same equivalence class (set) or not.

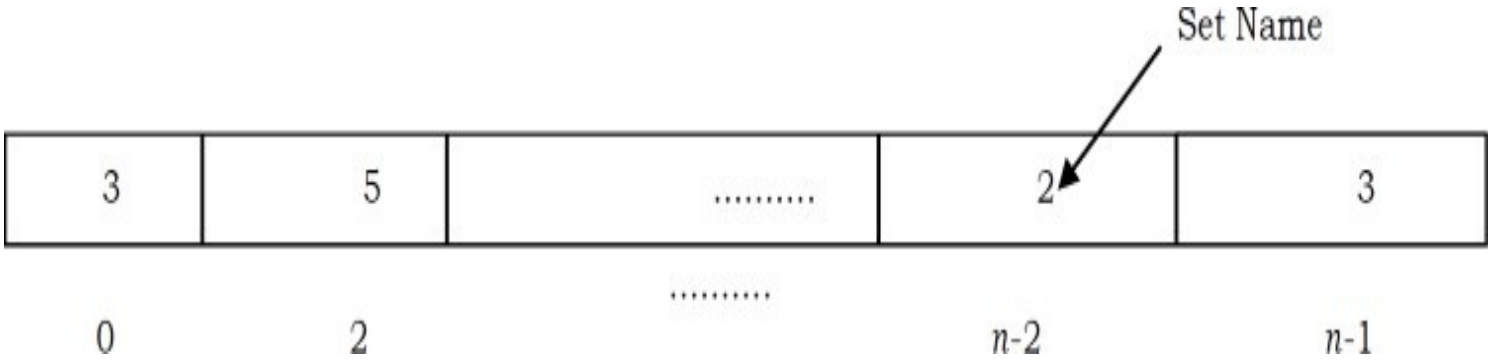
If they are not, then we apply UNION. This operation merges the two equivalence classes containing a and b into a new equivalence class by creating a new set $S_k = S_i \cup S_j$ and deletes S_i and S_j. Basically there are two ways to implement the above FIND/UNION operations:

- Fast FIND implementation (also called Quick FIND)
- Fast UNION operation implementation (also called Quick UNION)

8.6 Fast FIND Implementation (Quick FIND)

In this method, we use an array. As an example, in the representation below the array contains the set name for each element. For simplicity, let us assume that all the elements are numbered sequentially from 0 to $n - 1$.

In the example below, element 0 has the set name 3, element 1 has the set name 5, and so on. With this representation FIND takes only O(1) since for any element we can find the set name by accessing its array location in constant time.



In this representation, to perform UNION(*a*, *b*) [assuming that *a* is in set *i* and *b* is in set *j*] we need to scan the complete array and change all *i's to j*. This takes O(*n*).

A sequence of $n - 1$ unions take $O(n^2)$ time in the worst case. If there are $O(n^2)$ FIND operations, this performance is fine, as the average time complexity is O(1) for each UNION or FIND operation. If there are fewer FINDs, this complexity is not acceptable.

8.7 Fast UNION Implementation (Quick UNION)

In this and subsequent sections, we will discuss the faster *UNION* implementations and its variants. There are different ways of implementing this approach and the following is a list of a few of them.

- Fast UNION implementations (Slow FIND)
- Fast UNION implementations (Quick FIND)
- Fast UNION implementations with path compression

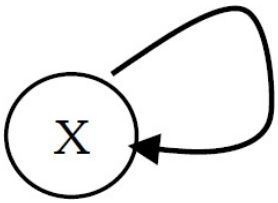
8.8 Fast UNION Implementation (Slow FIND)

As we have discussed, FIND operation returns the same answer (set name) if and only if they are in the same set. In representing disjoint sets, our main objective is to give a different set name for each group. In general we do not care about the name of the set. One possibility for implementing the set is *tree* as each element has only one *root* and we can use it as the set name.

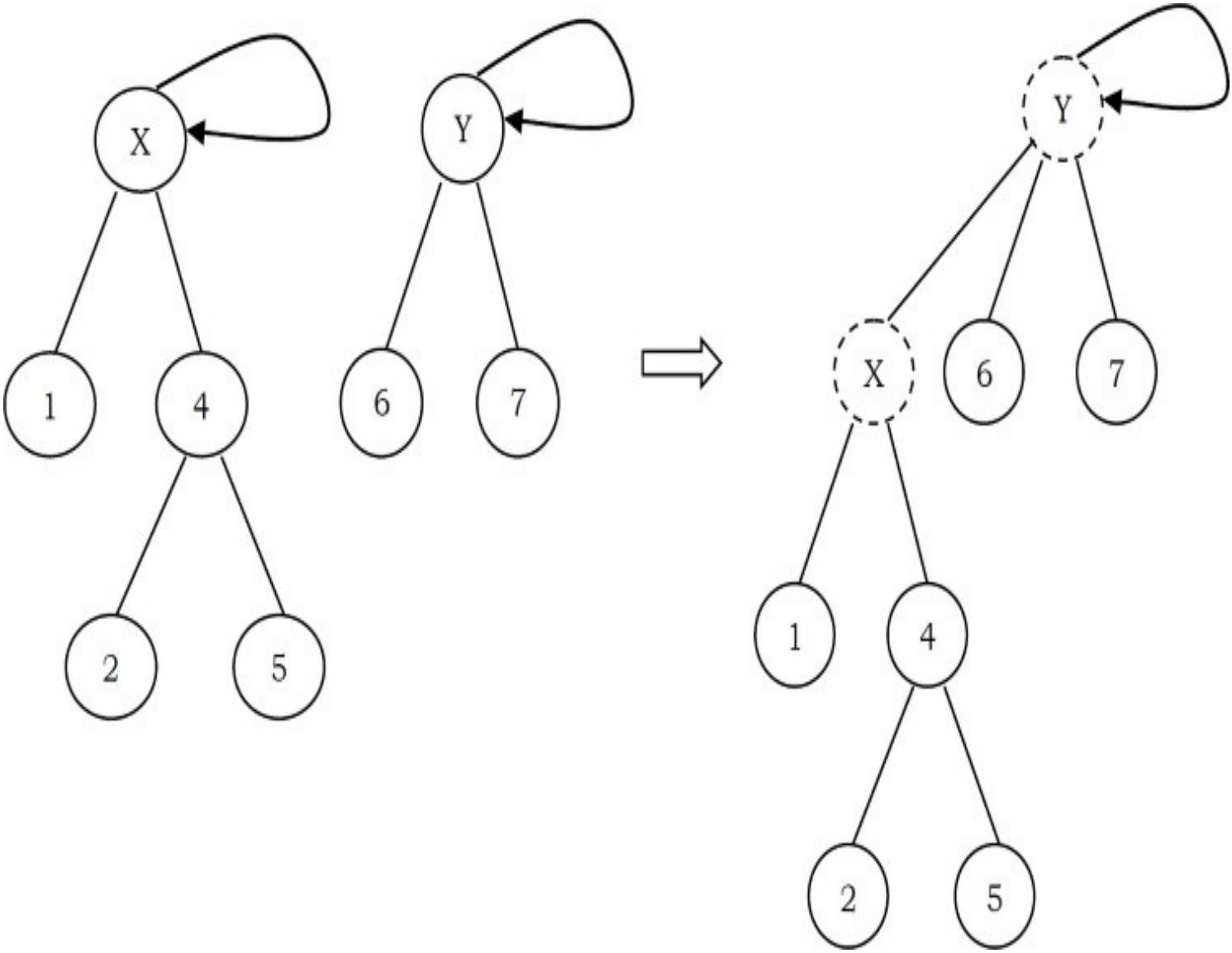
How are these represented? One possibility is using an array: for each element keep the *root* as its set name. But with this representation, we will have the same problem as that of FIND array implementation. To solve this problem, instead of storing the *root* we can keep the parent of the element. Therefore, using an array which stores the parent of each element solves our problem.

To differentiate the root node, let us assume its parent is the same as that of the element in the array. Based on this representation, MAKESET, FIND, UNION operations can be defined as:

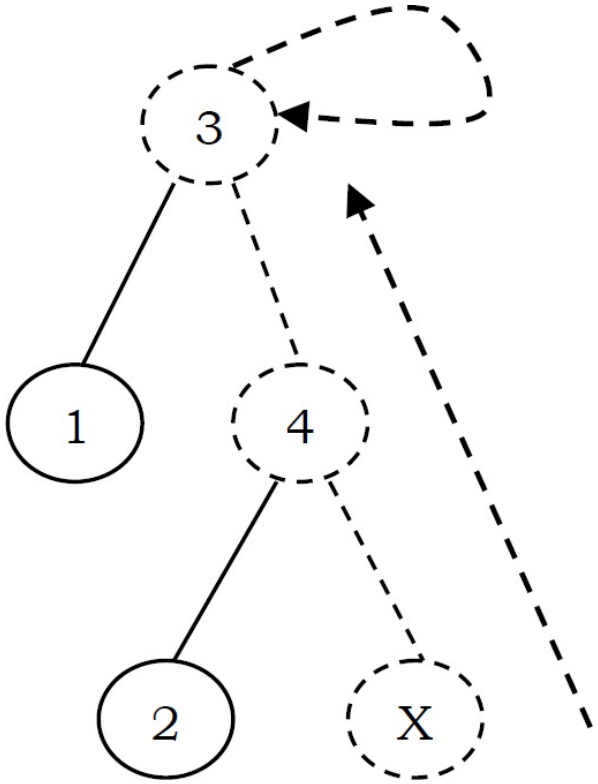
- (*X*): Creates a new set containing a single element *X* and in the array update the parent of *X* as *X*. That means root (set name) of *X* is *X*.



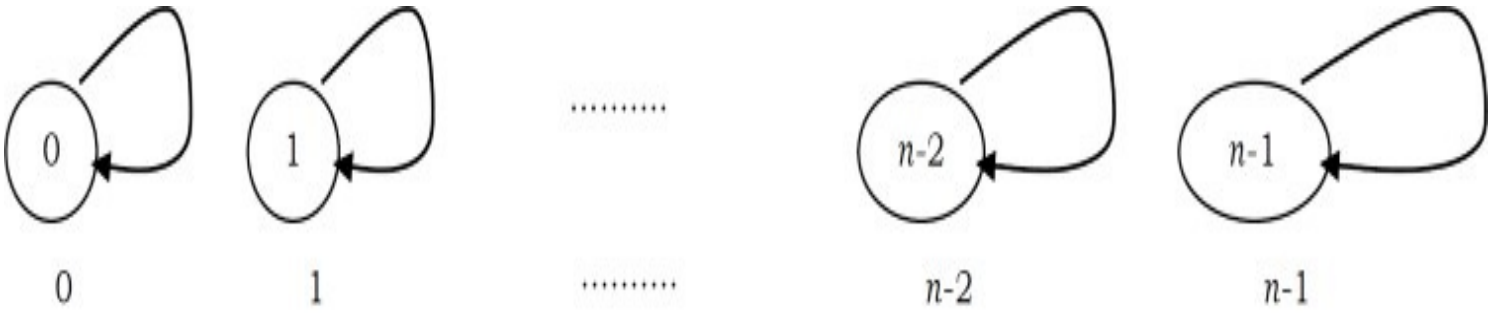
- UNION(X, *Y*): Replaces the two sets containing *X* and *Y* by their union and in the array updates the parent of *X* as *Y*.



- FIND(X): Returns the name of the set containing the element *X*. We keep on searching for *X's* set name until we come to the root of the tree.



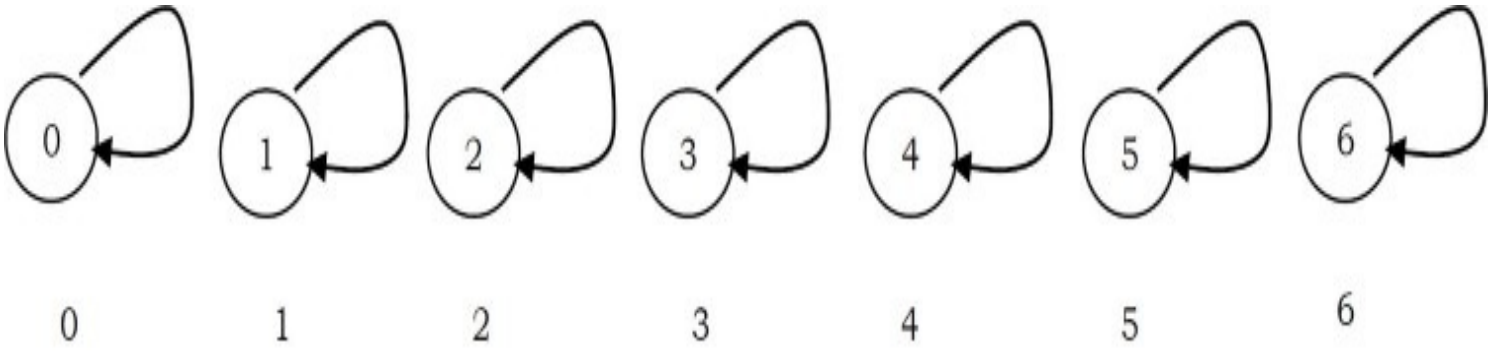
For the elements 0 to $n - 1$ the initial representation is:



Parent Array

To perform a UNION on two sets, we merge the two trees by making the root of one tree point to the root of the other.

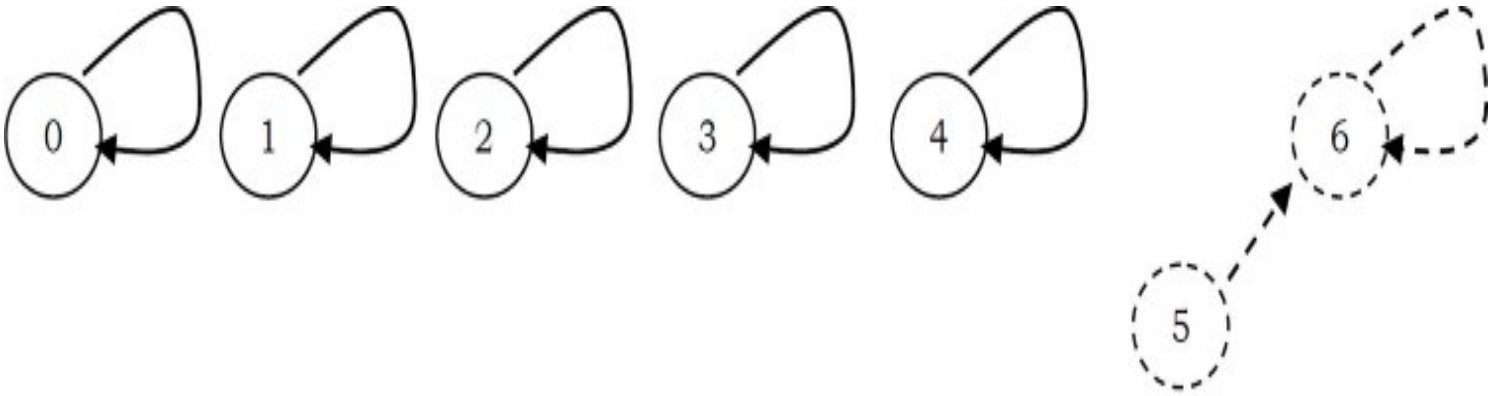
Initial Configuration for the elements 0 to 6



0	1	2	3	4	5	6

Parent Array

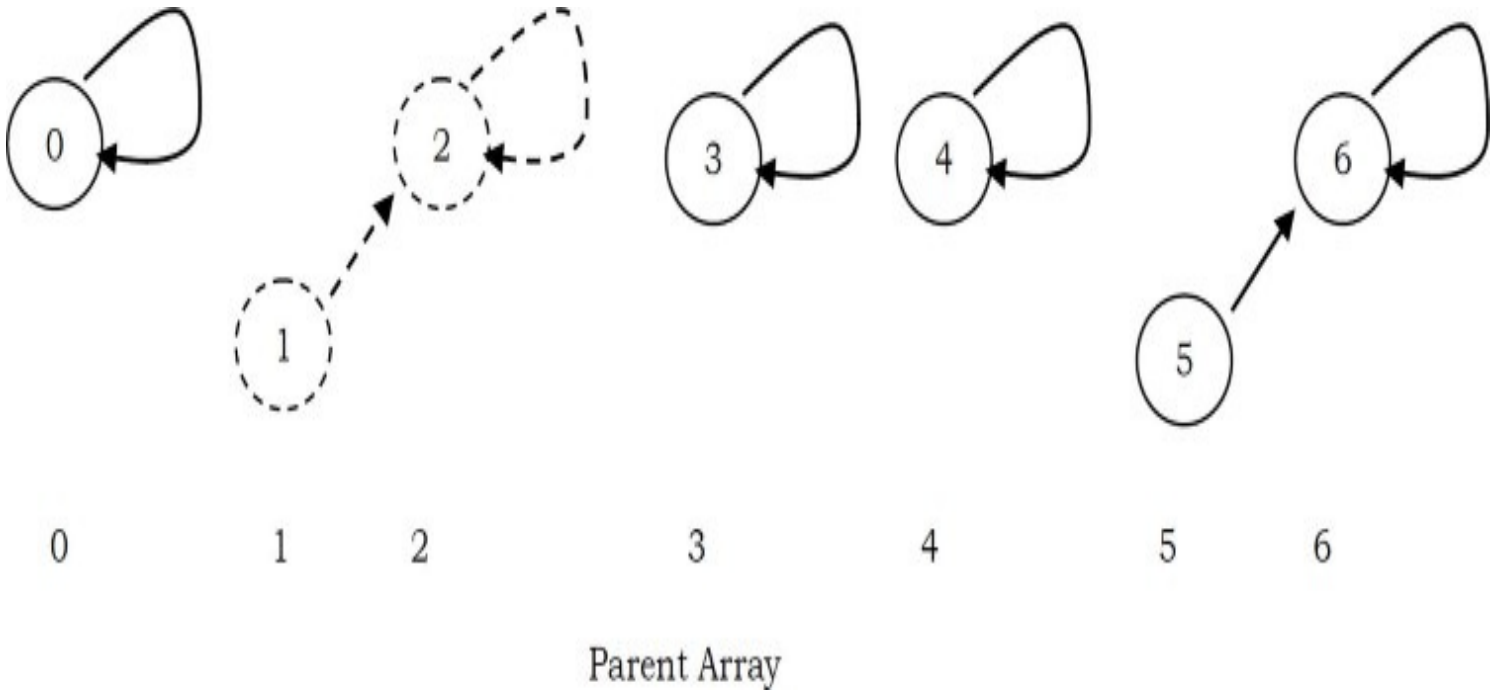
After UNION(5,6)



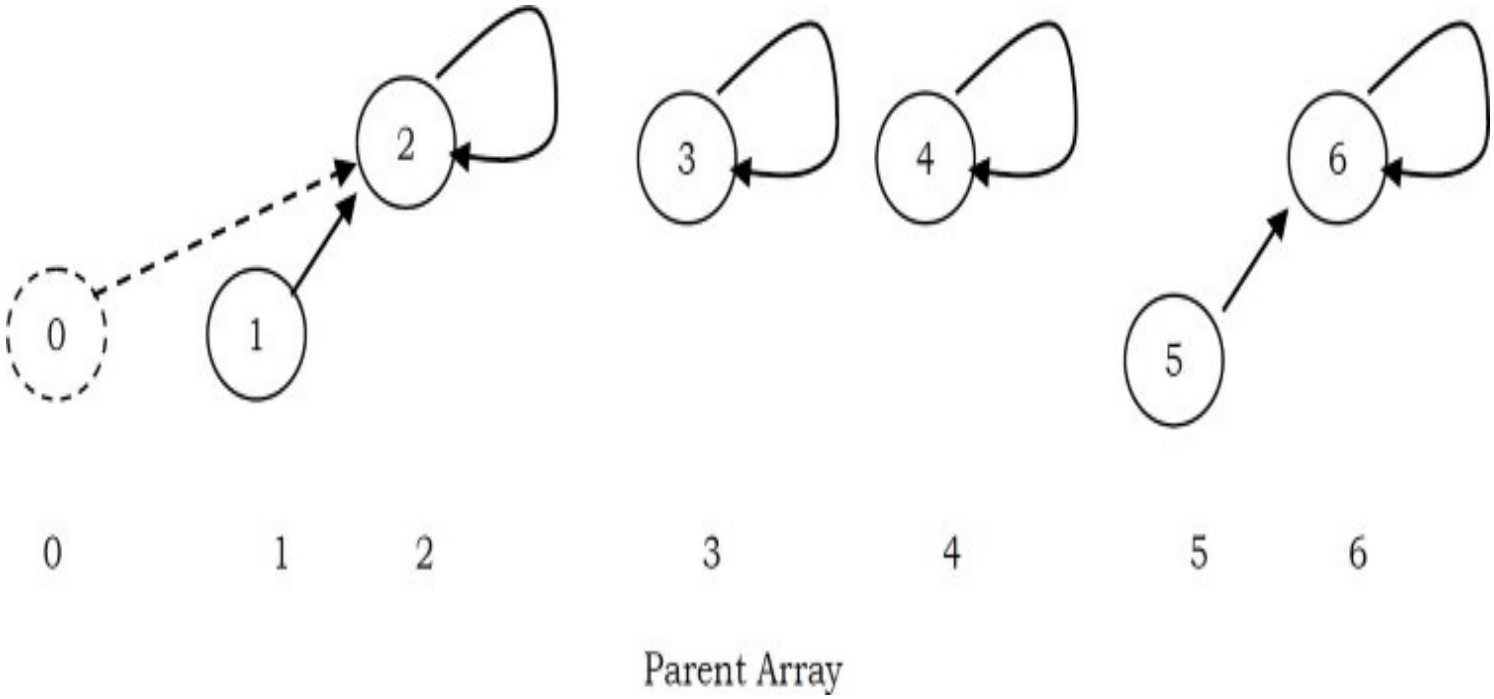
0	1	2	3	4	5	6

Parent Array

After UNION(1,2)



After UNION(0,2)



One important thing to observe here is, UNION operation is changing the root's parent only, but not for all the elements in the sets. Due to this, the time complexity of UNION operation is O(1).

A FIND(X) on element *X* is performed by returning the root of the tree containing *X*. The time to perform this operation is proportional to the depth of the node representing *X*.

Using this method, it is possible to create a tree of depth *n* - 1 (Skew Trees). The worst-case running time of a FIND is O(*n*) and *m* consecutive FIND operations take O(*mn*) time in the worst case.

MAKESET

```
void MAKESET( int S[], int size) {
    for(int  i = size-1; i >=0; i-- )
        S[i] = i;
}
```

FIND

```
int FIND(int S[], int size, int X) {
    if(!(X >= 0 && X < size))
        return -1;
    if( S[X] == X )
        return X;
    else  return FIND(S, S[X]);
}
```

UNION

```
void UNION( int S[], int size, int root1, int root2 ) {
    if(FIND(S, size, root1) == FIND(S, size, root2))
        return;
    if(!((root1 >= 0 && root1 < size) && (root2 >= 0 && root2 < size)))
        return;
    S[root1] = root2;
}
```

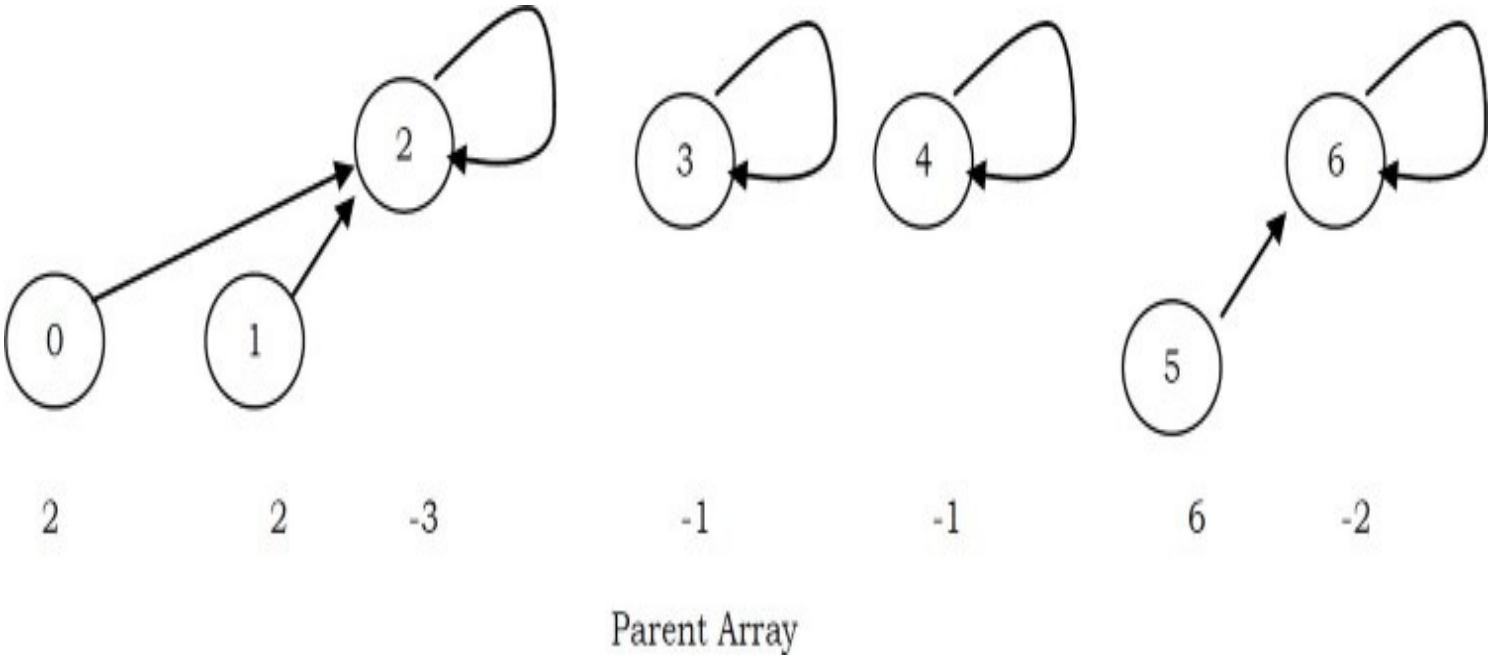
8.9 Fast UNION Implementations (Quick FIND)

The main problem with the previous approach is that, in the worst case we are getting the skew trees and as a result the FIND operation is taking O(n) time complexity. There are two ways to improve it:

- UNION by Size (also called UNION by Weight): Make the smaller tree a subtree of the larger tree
- UNION by Height (also called UNION by Rank): Make the tree with less height a subtree of the tree with more height

UNION by Size

In the earlier representation, for each element *i* we have stored *i* (in the parent array) for the root element and for other elements we have stored the parent of *i*. But in this approach we store negative of the size of the tree (that means, if the size of the tree is 3 then store –3 in the parent array for the root element). For the previous example (after UNION(0,2)), the new representation will look like:



Assume that the size of one element set is 1 and store – 1. Other than this there is no change.

MAKESET

```
void MAKESET( int S[], int size) {
    for(int  i = size-1; i >= 0; i-- )
        S[i] = -1;
}
```

FIND

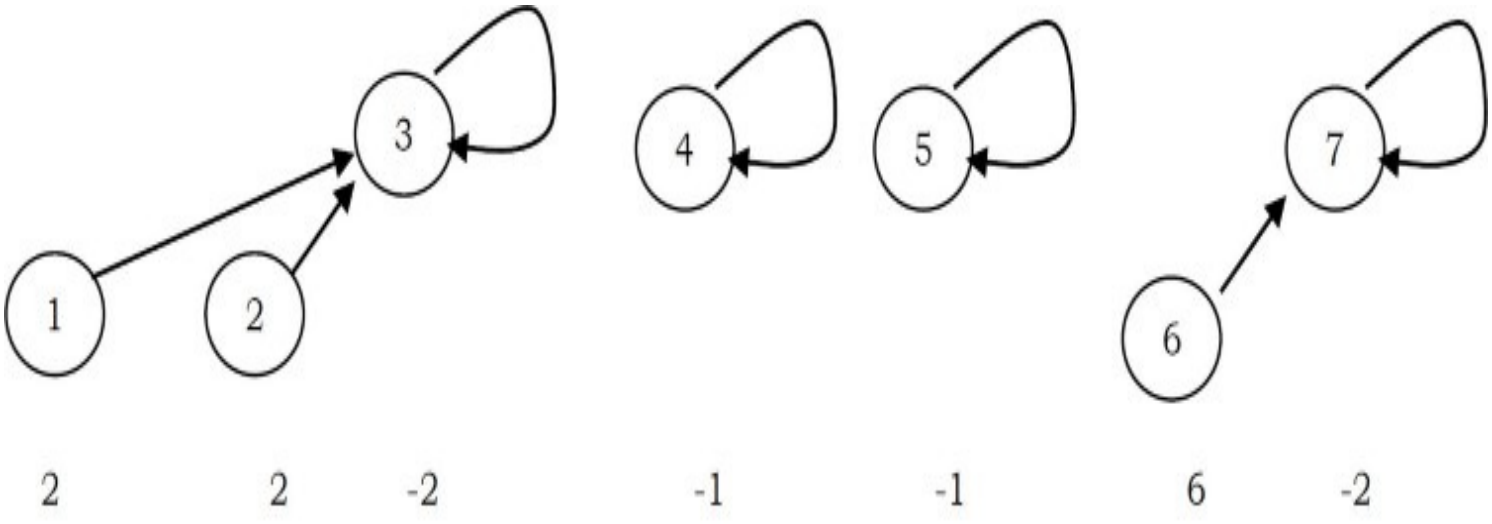
```
int FIND(int S[], int size, int X) {
    if(!(X >= 0 && X < size)) )
        return -1;
    if( S[X] == -1 )
        return X;
    else return FIND(S, S[X]);
}
```

UNION by Size

```
void UNIONBySize(int S[], int size, int root1, int root2) {
    if((FIND(S, size, root1) == FIND(S, size, root2)) && FIND(S, size, root1) != -1)
        return;
    if( S[root2] < S[root1] ) {
        S[root1] = root2;
        S[root2] += S[root1];
    }
    else {
        S[root2] = root1;
        S[root1] += S[root2];
    }
}
```

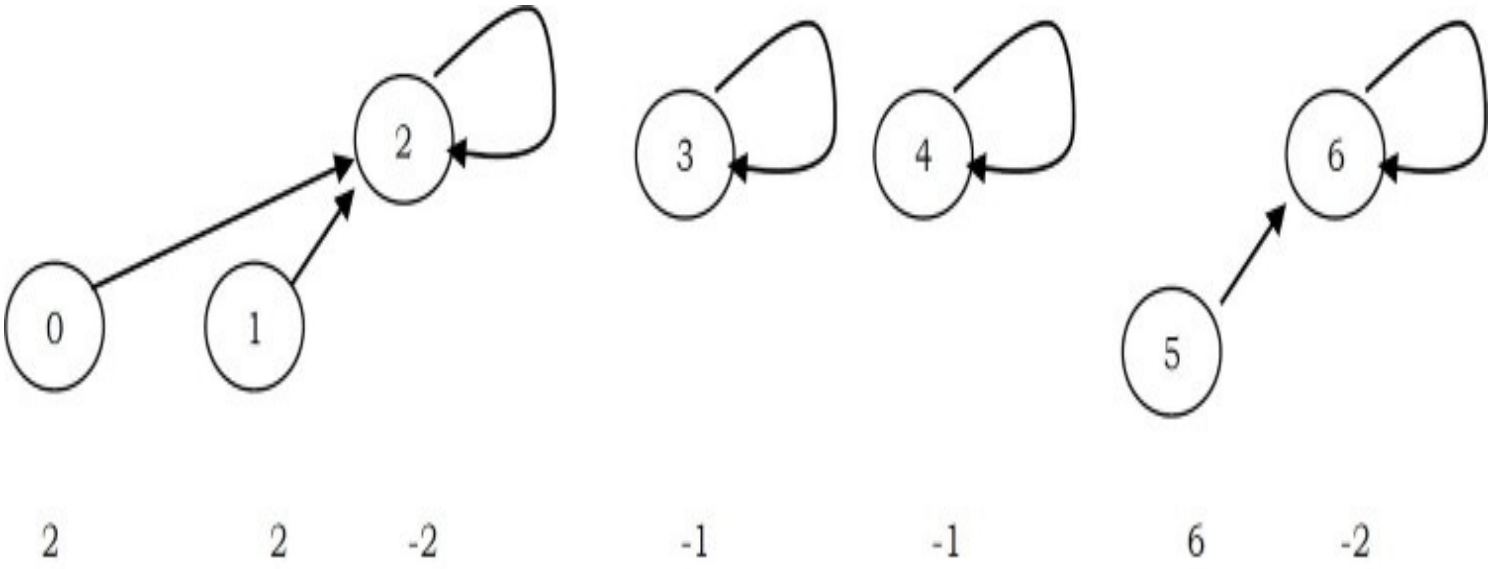
Note: There is no change in FIND operation implementation.

UNION by Height (UNION by Rank)



Parent Array

As in UNION by size, in this method we store negative of height of the tree (that means, if the height of the tree is 3 then we store –3 in the parent array for the root element). We assume the height of a tree with one element set is 1. For the previous example (after UNION(0,2)), the new representation will look like:



Parent Array

UNION by Height

```
void UNIONByHeight(int S[], int size, int root1, int root2) {
    if((FIND(S, size, root1) == FIND(S, size, root2)) && FIND(S, size, root1) != -1)
        return;
    if( S[root2] < S[root1] )
        S[root1] = root2;
    else {
        if( S[root2] == S[root1] ){
            S[root1]--;
            S[root2] = root1;
        }
    }
}
```

Note: For FIND operation there is no change in the implementation.

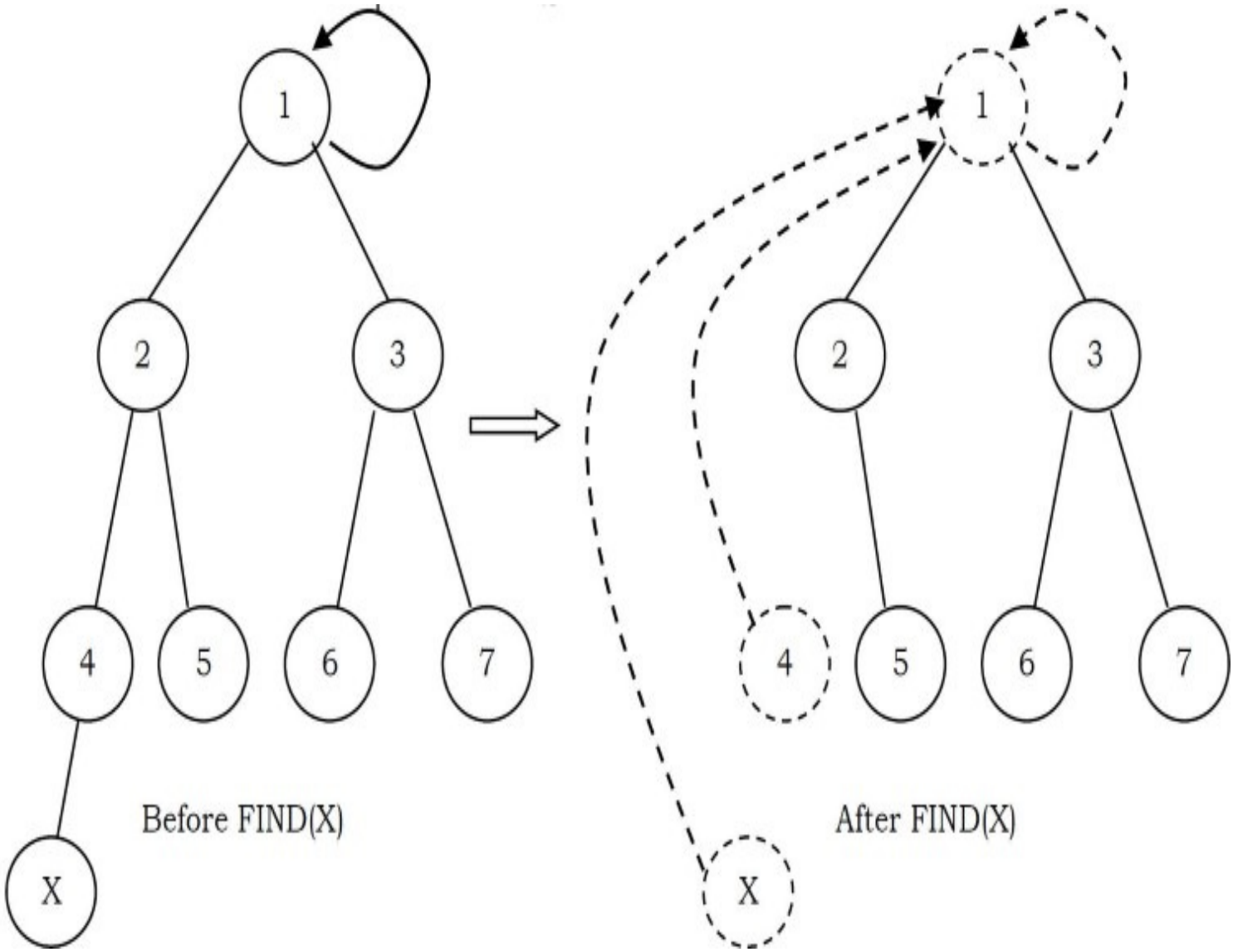
Comparing UNION by Size and UNION by Height

With UNION by size, the depth of any node is never more than *logn*. This is because a node is initially at depth 0. When its depth increases as a result of a UNION, it is placed in a tree that is at least twice as large as before. That means its depth can be increased at most *logn* times. This means that the running time for a FIND operation is O(*logn*), and a sequence of *m* operations takes O(*m logn*).

Similarly with UNION by height, if we take the UNION of two trees of the same height, the height of the UNION is one larger than the common height, and otherwise equal to the max of the two heights. This will keep the height of tree of *n* nodes from growing past O(*logn*). A sequence of *m* UNIONs and FINDs can then still cost O(*m logn*).

Path Compression

FIND operation traverses a list of nodes on the way to the root. We can make later FIND operations efficient by making each of these vertices point directly to the root. This process is called *path compression*. For example, in the FIND(*X*) operation, we travel from *X* to the root of the tree. The effect of path compression is that every node on the path from *X* to the root has its parent changed to the root.



With path compression the only change to the FIND function is that *S*[*X*] is made equal to the value returned by FIND. That means, after the root of the set is found recursively, *X* is made to point directly to it. This happen recursively to every node on the path to the root.

FIND with path compression

```
int FIND(int S[], int size, int X) {
    if(!(X >= 0 && X < size))
        return;
    if( S[X] <= 0 )
        return X;
    else return(S[X] = FIND( S, S[X]));
}
```

Note: Path compression is compatible with UNION by size but not with UNION by height as

there is no efficient way to change the height of the tree.

8.10 Summary

Performing *m* union-find operations on a set of *n* objects.

Algorithm	Worst-case time
Quick-Find	*mn*
Quick-Union	*mn*
Quick-Union by Size/Height	$n + m \log n$
Path compression	$n + m \log n$
Quick-Union by Size/Height + Path Compression	$(m + n) \log n$

8.11 Disjoint Sets: Problems & Solutions

Problem-1 Consider a list of cities $c_1, c_2,\ldots,c_n$. Assume that we have a relation R such that, for any i,j, $R(c_i,c_j)$ is 1 if cities c_i and c_j are in the same state, and 0 otherwise. If R is stored as a table, how much space does it require?

Solution: R must have an entry for every pair of cities. There are $\Theta(n^2)$ of these.

Problem-2 For Problem-1, using a Disjoint sets ADT, give an algorithm that puts each city in a set such that c_i and c_j are in the same set if and only if they are in the same state.

Solution:

```
for (i = 1; i<= n; i++) {
    MAKESET(ci);
    for (j = 1; j <= i-1; j++) {
        if(R(cj, ci)) {
            UNION(cj, ci);
            break;
        }
    }
}
```

Problem-3 For Problem-1, when the cities are stored in the Disjoint sets ADT, if we are given two cities c_i and c_j, how do we check if they are in the same state?

Solution: Cities c_i and c_j are in the same state if and only if FIND(c_i) = FIND(c_j).

Problem-4 For Problem-1, if we use linked-lists with UNION by size to implement the union-find ADT, how much space do we use to store the cities?

Solution: There is one node per city, so the space is $\Theta(n)$.

Problem-5 For Problem-1, if we use trees with UNION by rank, what is the worst-case running time of the algorithm from Problem-2?

Solution: Whenever we do a UNION in the algorithm from Problem-2, the second argument is a tree of size 1. Therefore, all trees have height 1, so each union takes time O(1). The worst-case running time is then $\Theta(n^2)$.

Problem-6 If we use trees without union-by-rank, what is the worst-case running time of the algorithm from Problem-2? Are there more worst-case scenarios than Problem-5?

Solution: Because of the special case of the unions, union-by-rank does not make a difference for our algorithm. Hence, everything is the same as in Problem-5.

Problem-7 With the quick-union algorithm we know that a sequence of n operations (*unions* and *finds*) can take slightly more than linear time in the worst case. Explain why if all the *finds* are done before all the *unions*, a sequence of n operations is guaranteed to take O(n) time.

Solution: If the *find* operations are performed first, then the *find* operations take O(1) time each because every item is the root of its own tree. No item has a parent, so finding the set an item is in takes a fixed number of operations. Union operations always take O(1) time. Hence, a sequence of n operations with all the *finds* before the *unions* takes O(n) time.

Problem-8 With reference to Problem-7, explain why if all the unions are done before all the finds, a sequence of n operations is guaranteed to take O(n) time.

Solution: This problem requires amortized analysis. *Find* operations can be expensive, but this expensive *find* operation is balanced out by lots of cheap *union* operations.

The accounting is as follows. *Union* operations always take O(1) time, so let's say they have an actual cost of $\sqrt{2}1$. Assign each *union* operation an amortized cost of $\sqrt{2}2$, so every *union* operation puts $\sqrt{2}1$ in the account. Each *union* operation creates a new child. (Some node that was not a child of any other node before is a child now.) When all the union operations are done, there is \$1 in the account for every child, or in other words, for every node with a depth of one or greater. Let's say that a *find*(u) operation costs $\sqrt{2}1$ if u is a root. For any other node, the *find* operation costs an additional $\sqrt{2}1$ for each parent pointer the *find* operation traverses. So the actual cost is $\sqrt{2}$ (1 + d), where d is the depth of u. Assign each *find* operation an amortized cost

of $\sqrt{2}2$. This covers the case where *u* is a root or a child of a root. For each additional parent pointer traversed, $\sqrt{2}1$ is withdrawn from the account to pay for it.

Fortunately, path compression changes the parent pointers of all the nodes we pay $\sqrt{2}1$ to traverse, so these nodes become children of the root. All of the traversed nodes whose depths are 2 or greater move up, so their depths are now 1. We will never have to pay to traverse these nodes again. Say that a node is a grandchild if its depth is 2 or greater.

Every time *find*(*u*) visits a grandchild, $\sqrt{2}1$ is withdrawn from the account, but the grandchild is no longer a grandchild. So the maximum number of dollars that can ever be withdrawn from the account is the number of grandchildren. But we initially put $1 in the bank for every child, and every grandchild is a child, so the bank balance will never drop below zero. Therefore, the amortization works out. *Union* and *find* operations both have amortized costs of $\sqrt{2}2$, so any sequence of n operations where all the unions are done first takes O(*n*) time.

CHAPTER

9

GRAPH ALGORITHMS

9.1 Introduction

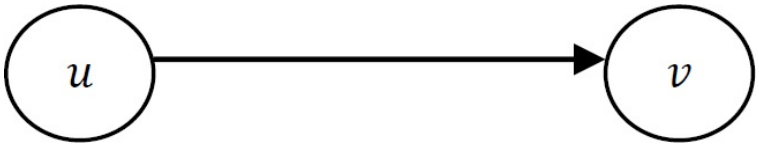
In the real world, many problems are represented in terms of objects and connections between them. For example, in an airline route map, we might be interested in questions like: “What’s the fastest way to go from Hyderabad to New York?” *or* “What is the cheapest way to go from Hyderabad to New York?” To answer these questions we need information about connections (airline routes) between objects (towns). Graphs are data structures used for solving these kinds of problems.

9.2 Glossary

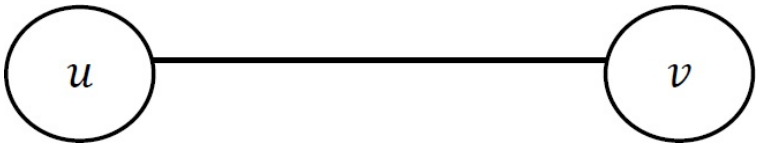
Graph: A graph is a pair (*V, E*), where *V* is a set of nodes, called *vertices,* and £ is a collection of pairs of vertices, called *edges*.

- *Vertices* and *edges* are positions and store elements

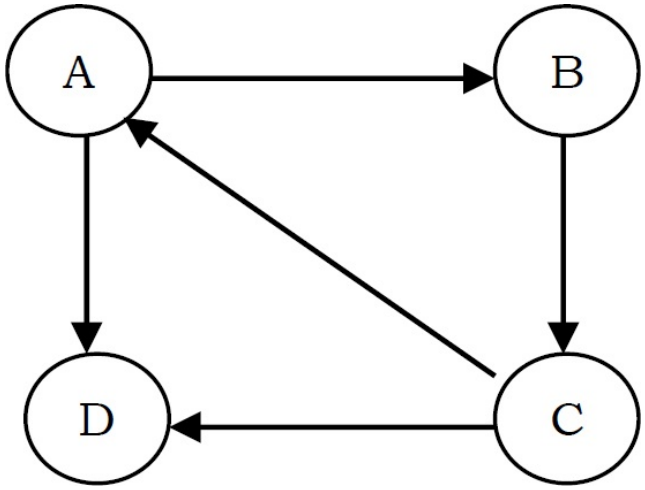
- Definitions that we use:
  - *Directed edge:*
    - ordered pair of vertices (u, v)
    - first vertex u is the origin
    - second vertex v is the destination
    - Example: one-way road traffic



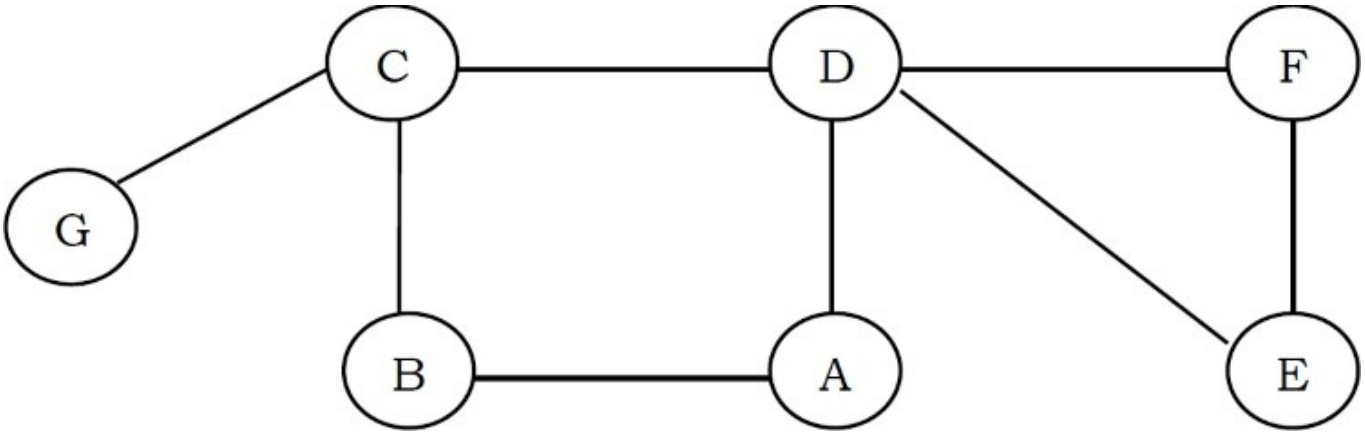
  - *Undirected edge:*
    - unordered pair of vertices (u, v)
    - Example: railway lines



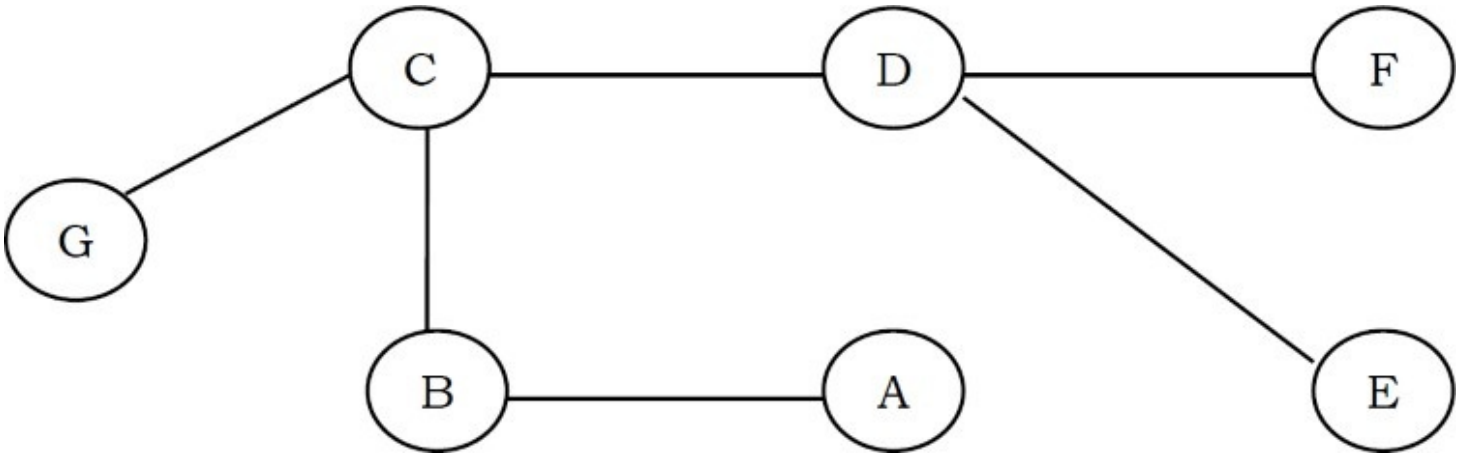
  - *Directed graph:*
    - all the edges are directed
    - Example: route network



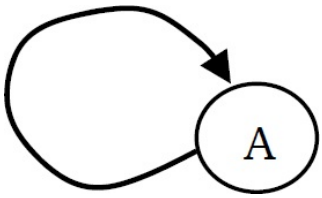
  - *Undirected graph:*
    - all the edges are undirected
    - Example: flight network



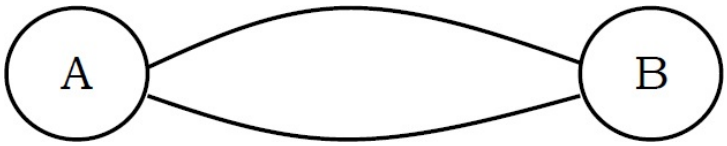
- When an edge connects two vertices, the vertices are said to be adjacent to each other and the edge is incident on both vertices.
- A graph with no cycles is called a *tree*. A tree is an acyclic connected graph.



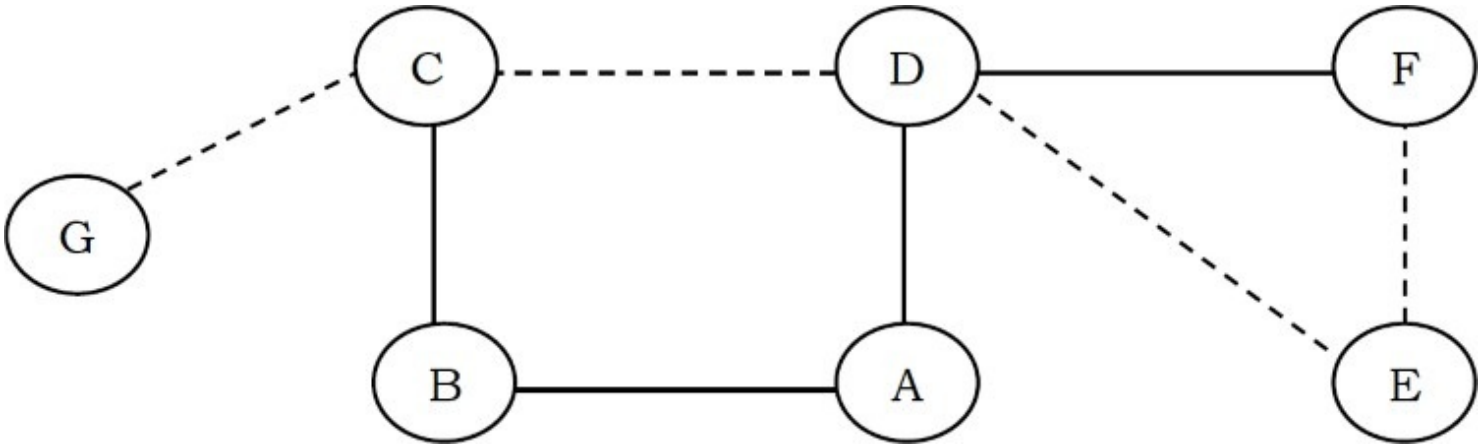
- A self loop is an edge that connects a vertex to itself.



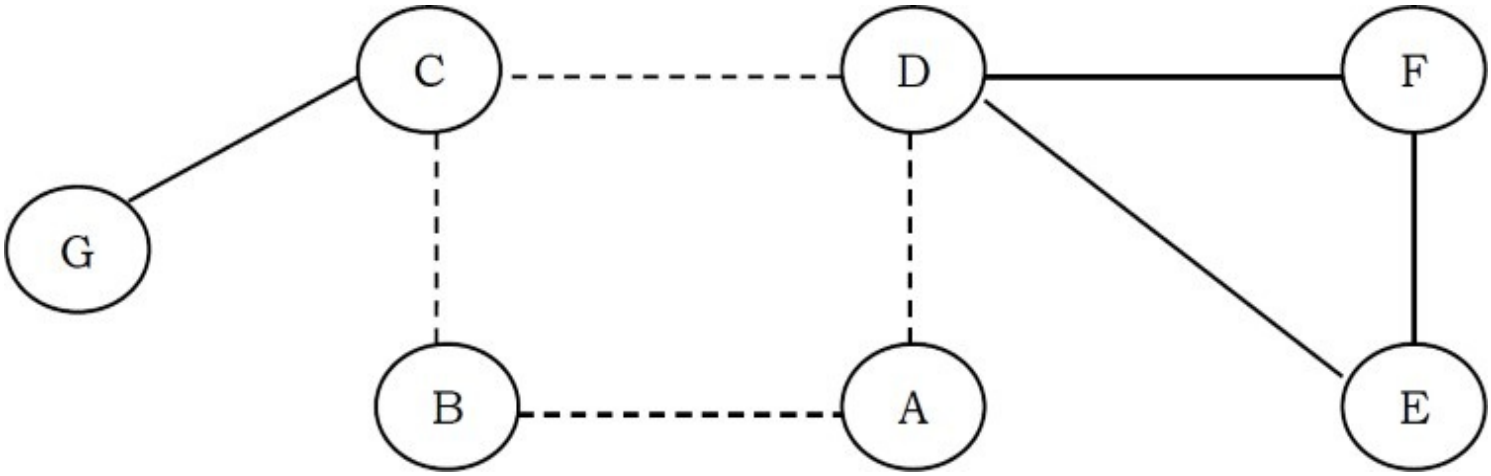
- Two edges are parallel if they connect the same pair of vertices.



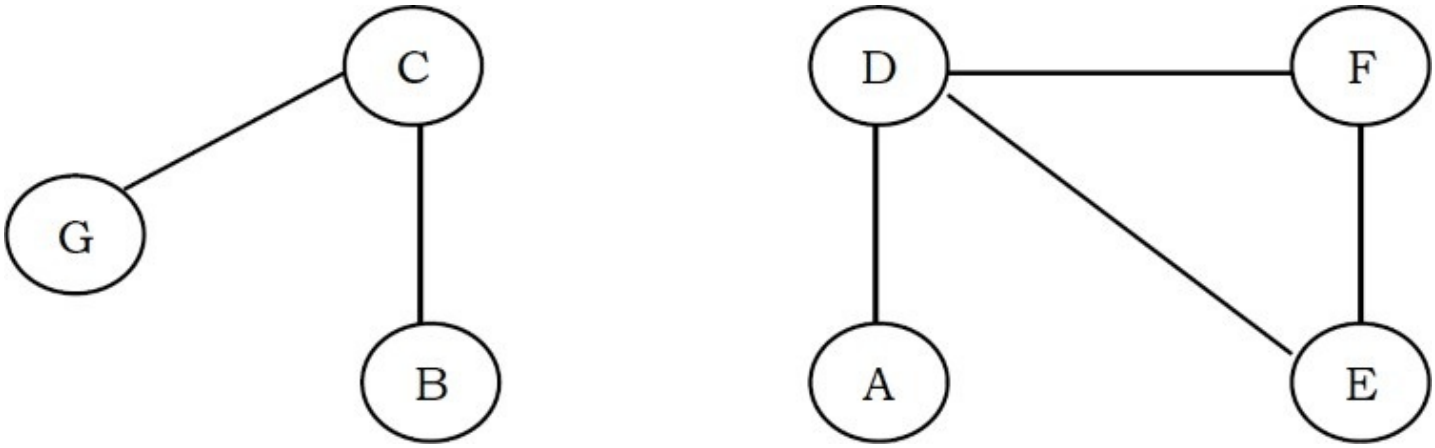
- The *D*egree of a vertex is the number of edges incident on it.
- A subgraph is a subset of a graph's edges (with associated vertices) that form a graph.
- A path in a graph is a sequence of adjacent vertices. *S*imple path is a path with no repeated vertices. In the graph below, the dotted lines represent a path from *G* to *E*.



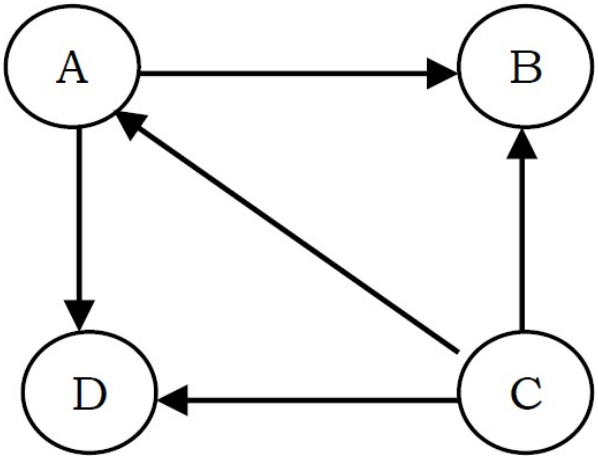
- A cycle is a path where the first and last vertices are the same. A simple cycle is a cycle with no repeated vertices or edges (except the first and last vertices).



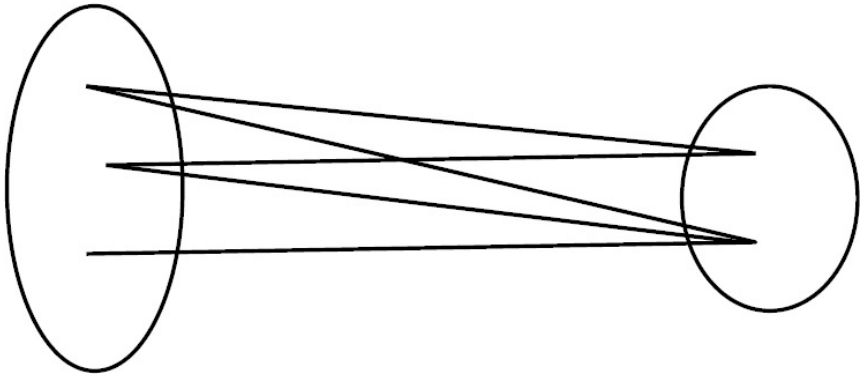
- We say that one vertex is connected to another if there is a path that contains both of them.
- A graph is connected if there is a path from *every* vertex to every other vertex.
- If a graph is not connected then it consists of a set of connected components.



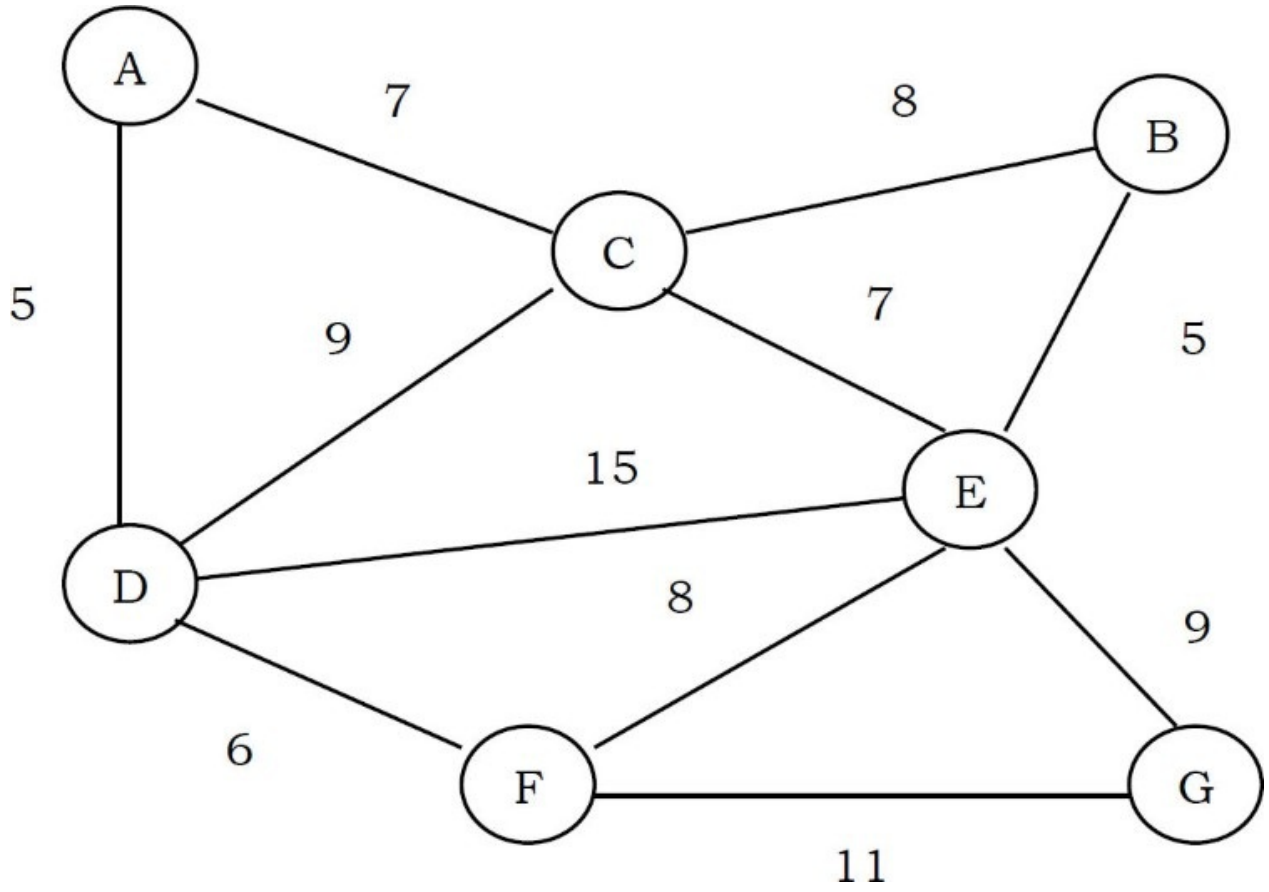
- A *directed* acyclic graph [DAG] is a directed graph with no cycles.



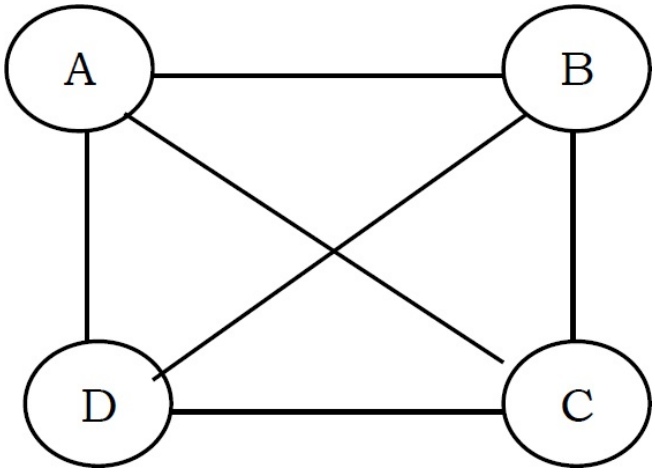
- A forest is a disjoint set of trees.
- A spanning tree of a connected graph is a subgraph that contains all of that graph's vertices and is a single tree. A spanning forest of a graph is the union of spanning trees of its connected components.
- A bipartite graph is a graph whose vertices can be divided into two sets such that all edges connect a vertex in one set with a vertex in the other set.



- In *weighted graphs* integers (*weights*) are assigned to each edge to represent (distances or costs).



- Graphs with all edges present are called *complete* graphs.



- Graphs with relatively few edges (generally if it edges < $|V|$ log $|V|$) are called *sparse graphs*.
- Graphs with relatively few of the possible edges missing are called *dense*.
- Directed weighted graphs are sometimes called *network*.
- We will denote the number of vertices in a given graph by $|V|$, and the number of edges by $|E|$. Note that E can range anywhere from 0 to $|V|(|V| - 1)/2$ (in undirected graph). This is because each node can connect to every other node.

9.3 Applications of Graphs

- Representing relationships between components in electronic circuits
- Transportation networks: Highway network, Flight network
- Computer networks: Local area network, Internet, Web
- Databases: For representing ER (Entity Relationship) diagrams in databases, for representing dependency of tables in databases

9.4 Graph Representation

As in other ADTs, to manipulate graphs we need to represent them in some useful form. Basically, there are three ways of doing this:

- Adjacency Matrix
- Adjacency List
- Adjacency Set

Adjacency Matrix

Graph Declaration for Adjacency Matrix

First, let us look at the components of the graph data structure. To represent graphs, we need the number of vertices, the number of edges and also their interconnections. So, the graph can be declared as:

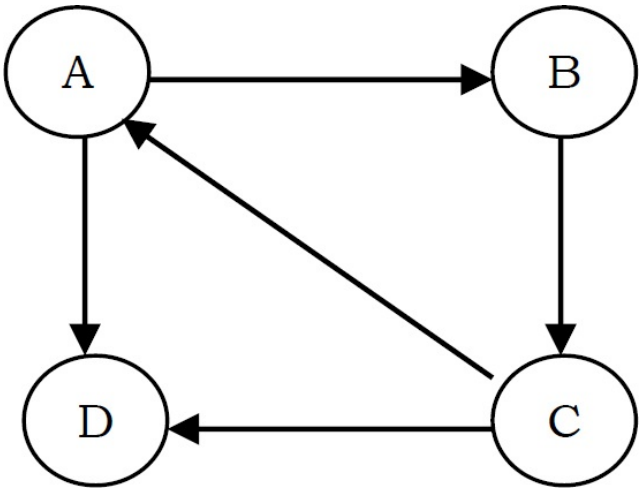
```
struct Graph {
  int V;
  int E;
  int **Adj; //Since we need two dimensional matrix
};
```

Description

In this method, we use a matrix with size $V \times V$. The values of matrix are boolean. Let us assume the matrix is *Adj*. The value *Adj*[*u, v*] is set to 1 if there is an edge from vertex u to vertex v and 0 otherwise.

In the matrix, each edge is represented by two bits for undirected graphs. That means, an edge from **u** to **v** is represented by 1 value in both *Adj*[**u**,**v**] and *Adj*[*u,v*]. To save time, we can process only half of this symmetric matrix. Also, we can assume that there is an “edge” from each vertex to itself. So, *Adj*[u, u] is set to 1 for all vertices.

If the graph is a directed graph then we need to mark only one entry in the adjacency matrix. As an example, consider the directed graph below.



The adjacency matrix for this graph can be given as:

	A	B	C	D
A	0	1	0	1
B	0	0	1	0
C	1	0	0	1
D	0	0	0	0

Now, let us concentrate on the implementation. To read a graph, one way is to first read the vertex names and then read pairs of vertex names (edges). The code below reads an undirected graph.

```
//This code creates a graph with adj matrix representation
struct Graph *adjMatrixOfGraph() {
    int i, u, v;
    struct Graph *G = (struct Graph *) malloc(sizeof(struct Graph));
    if(!G) {
        printf("Memory Error");
        return;
    }
    scanf("Number of Vertices: %d, Number of Edges:%d", &G→V, &G→E);
    G→Adj = malloc(sizeof(G→V  * G→V));
    for(u = 0; u < G→V; u++)
            for(v = 0; v < G→V; v++)
                    G→Adj[v][v] = 0;
    for(i = 0;  i < G→E; i++) {
        //Read an edge
        scanf("Reading Edge: %d %d", &u, &v);
        //For undirected graphs set both the bits
        G→ Adj[u][v] = 1;
        G→ Adj[v][u] = 1;
    }
    return G;
}
```

The adjacency matrix representation is good if the graphs are dense. The matrix requires $O(V^2)$ bits of storage and $O(V^2)$ time for initialization. If the number of edges is proportional to V^2, then there is no problem because V^2 steps are required to read the edges. If the graph is sparse, the initialization of the matrix dominates the running time of the algorithm as it takes takes $O(V^2)$.

Adjacency List

Graph Declaration for Adjacency List

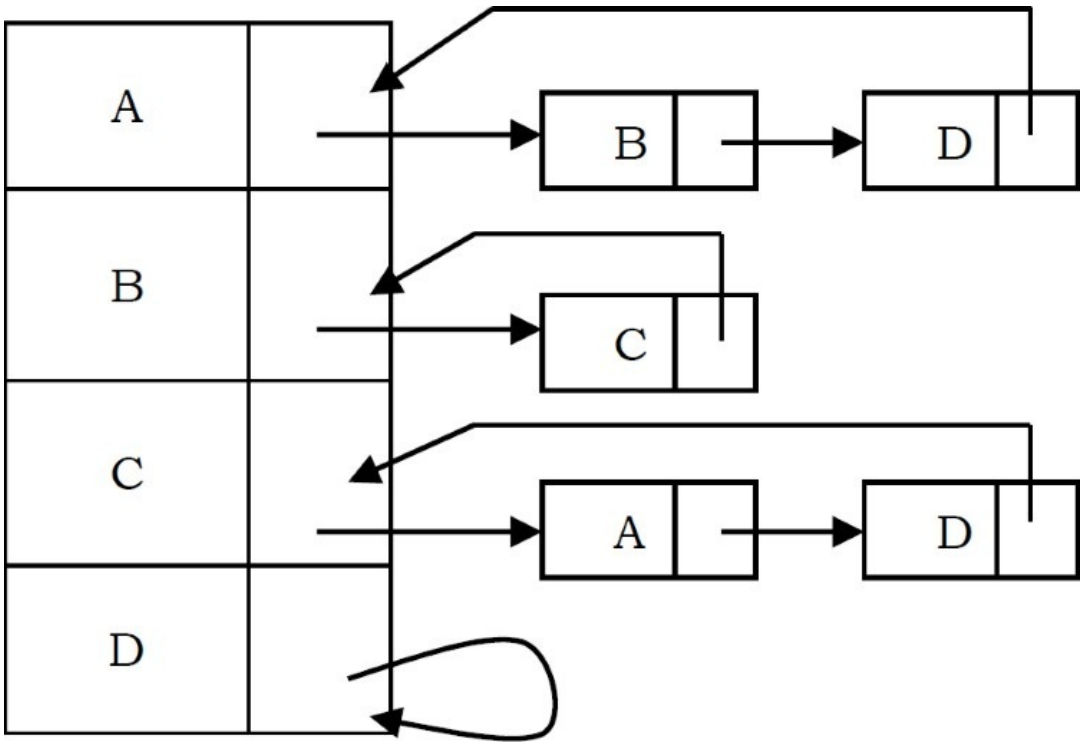
In this representation all the vertices connected to a vertex *v* are listed on an adjacency list for that vertex *v*. This can be easily implemented with linked lists. That means, for each vertex *v* we use a linked list and list nodes represents the connections between *v* and other vertices to which *v* has an edge.

The total number of linked lists is equal to the number of vertices in the graph. The graph ADT can be declared as:

```
struct Graph {
    int V;
    int E;
    int *Adj; //head pointers to linked list
};
```

Description

Considering the same example as that of the adjacency matrix, the adjacency list representation can be given as:



Since vertex A has an edge for B and D, we have added them in the adjacency list for A. The same is the case with other vertices as well.

```
//Nodes of the Linked List
struct ListNode {
    int vertexNumber;
    struct ListNode *next;
}

//This code creates a graph with adj list representation
struct Graph *adjListOfGraph() {
    int i, x, y;
    struct ListNode *temp;
    struct Graph *G = (struct Graph *) malloc(sizeof(struct Graph));
    if(!G) {
        printf("Memory Error");
        return;
    }
    scanf("Number of Vertices: %d, Number of Edges:%d", &G→V, &G→E);
    G→Adj = malloc(G→V * sizeof(struct ListNode));

    for(i = 0; i  < G→V; i++) {
        G→Adj[i] = (struct ListNode *) malloc(sizeof(struct ListNode));
        G→Adj[i]→vertexNumber = i;
        G→Adj[i]→next = G→ Adj[i];
    }
    for(i = 0;  i < E; i++) {
        //Read an edge
        scanf("Reading Edge: %d %d", &x, &y);
        temp = (struct ListNode *) malloc(struct ListNode);
        temp→vertexNumber = y;
        temp→next = G→Adj[x];
        G→ Adj[x]→next = temp;
        temp = (struct ListNode *) malloc(struct ListNode);
        temp→vertexNumber = y;
        temp→next = G→ Adj[y];
        G→Adj[y]→ next= temp;
    }

    retutn G;
}
```

For this representation, the order of edges in the input is *important*. This is because they determine the order of the vertices on the adjacency lists. The same graph can be represented in many different ways in an adjacency list. The order in which edges appear on the adjacency list affects the order in which edges are processed by algorithms.

Disadvantages of Adjacency Lists

Using adjacency list representation we cannot perform some operations efficiently. As an example, consider the case of deleting a node. . In adjacency list representation, it is not enugh if we simply delete a node from the list representation, if we delete a node from the adjacency list then that is enough. For each node on the adjacency list of that node specifies another vertex. We need to search other nodes linked list also for deleting it. This problem can be solved by linking the two list nodes that correspond to a particular edge and making the adjacency lists doubly linked. But all these extra links are risky to process.

Adjacency Set

It is very much similar to adjacency list but instead of using Linked lists, Disjoint Sets [Union-Find] are used. For more details refer to the *Disjoint Sets ADT* chapter.

Comparison of Graph Representations

Directed and undirected graphs are represented with the same structures. For directed graphs, everything is the same, except that each edge is represented just once. An edge from x to y is represented by a 1 value in $Agj[x][y]$ in the adjacency matrix, or by adding y on x's adjacency list. For weighted graphs, everything is the same, except fill the adjacency matrix with weights instead of boolean values.

Representation	Space	Checking edge between v and w?	Iterate over edges incident to v?
List of edges	E	E	E
Adj Matrix	V^2	1	V
Adj List	$E+V$	$Degree(v)$	$Degree(v)$
Adj Set	$E+V$	$log(Degree(v))$	$Degree(v)$

9.5 Graph Traversals

To solve problems on graphs, we need a mechanism for traversing the graphs. Graph traversal algorithms are also called *graph search* algorithms. Like trees traversal algorithms (Inorder, Preorder, Postorder and Level-Order traversals), graph search algorithms can be thought of as starting at some source vertex in a graph and "searching" the graph by going through the edges and marking the vertices. Now, we will discuss two such algorithms for traversing the graphs.

- Depth First Search [DFS]
- Breadth First Search [BFS]

Depth First Search [DFS]

DFS algorithm works in a manner similar to preorder traversal of the trees. Like preorder traversal, internally this algorithm also uses stack.

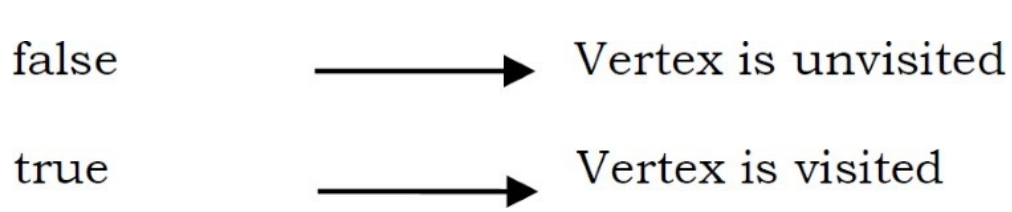
Let us consider the following example. Suppose a person is trapped inside a maze. To come out from that maze, the person visits each path and each intersection (in the worst case). Let us say the person uses two colors of paint to mark the intersections already passed. When discovering a new intersection, it is marked grey, and he continues to go deeper.

After reaching a "dead end" the person knows that there is no more unexplored path from the grey intersection, which now is completed, and he marks it with black. This "dead end" is either an intersection which has already been marked grey or black, or simply a path that does not lead to an intersection.

The intersections of the maze are the vertices and the paths between the intersections are the edges of the graph. The process of returning from the "dead end" is called *backtracking*. We are trying to go away from the starting vertex into the graph as deep as possible, until we have to backtrack to the preceding grey vertex. In DFS algorithm, we encounter the following types of edges.

Tree edge: encounter new vertex
Back edge: from descendent to ancestor
Forward edge: from ancestor to descendent
Cross edge: between a tree or subtrees

For most algorithms boolean classification, unvisited/visited is enough (for three color implementation refer to problems section). That means, for some problems we need to use three colors, but for our discussion two colors are enough.



Initially all vertices are marked unvisited (false). The DFS algorithm starts at a vertex *u* in the graph. By starting at vertex *u* it considers the edges from *u* to other vertices. If the edge leads to an already visited vertex, then backtrack to current vertex *u*. If an edge leads to an unvisited vertex, then go to that vertex and start processing from that vertex. That means the new vertex becomes the current vertex. Follow this process until we reach the dead-end. At this point start *backtracking*.

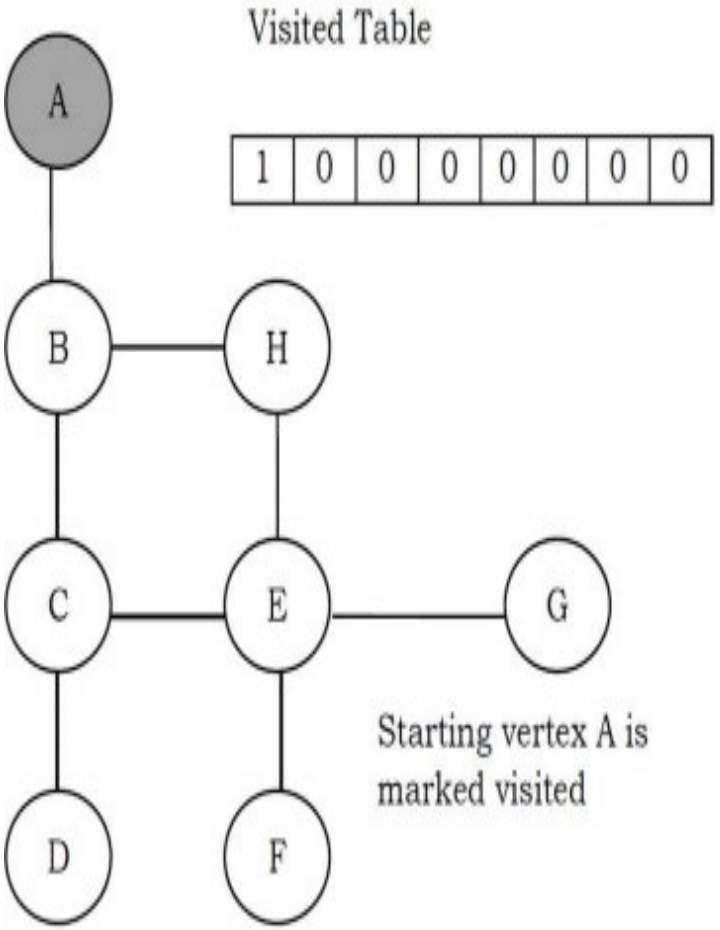
The process terminates when backtracking leads back to the start vertex. The algorithm based on this mechanism is given below: assume Visited[] is a global array.

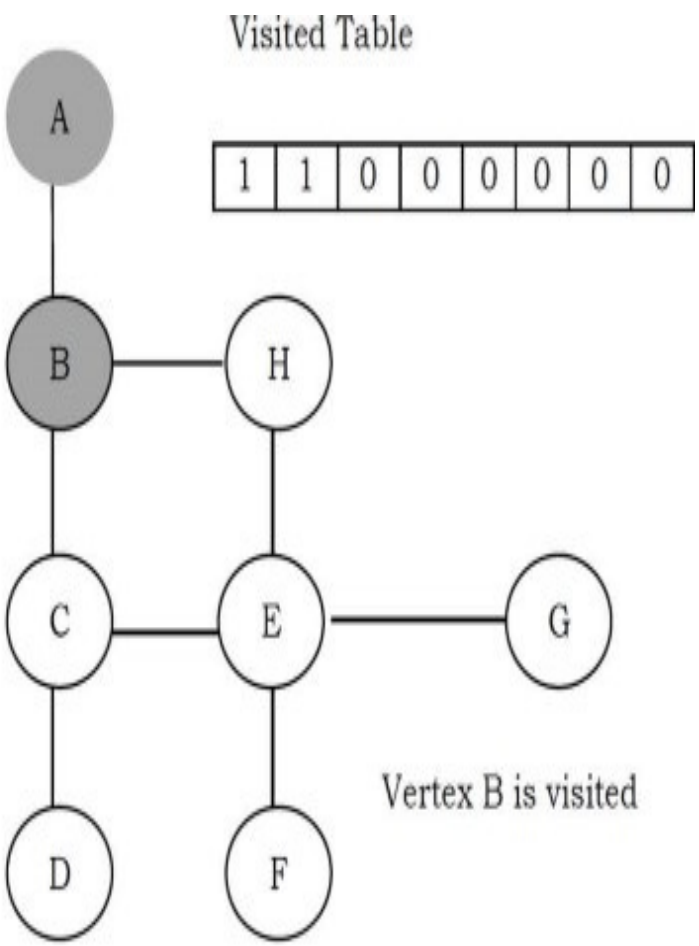
```
int Visited[G→V];
void DFS(struct Graph *G, int u) {
    Visited[u] = 1;
    for( int v = 0; v < G→V; v++ ) {

        /* For example, if the adjacency matrix is used for representing the
           graph, then the condition to be used for finding unvisited adjacent
           vertex of u  is: if( !Visited[v] && G→Adj[u][v] ) */

        for each unvisited adjacent node v of u {
            DFS(G, v);
        }
    }
}
void DFSTraversal(struct Graph *G) {
    for (int i = 0; i< G→V;i++)
        Visited[i]=0;

    //This loop is required if the graph has more than one component
    for (int i = 0; i< G→V;i++)
        if(!Visited[i])
            DFS(G, i);
}
```

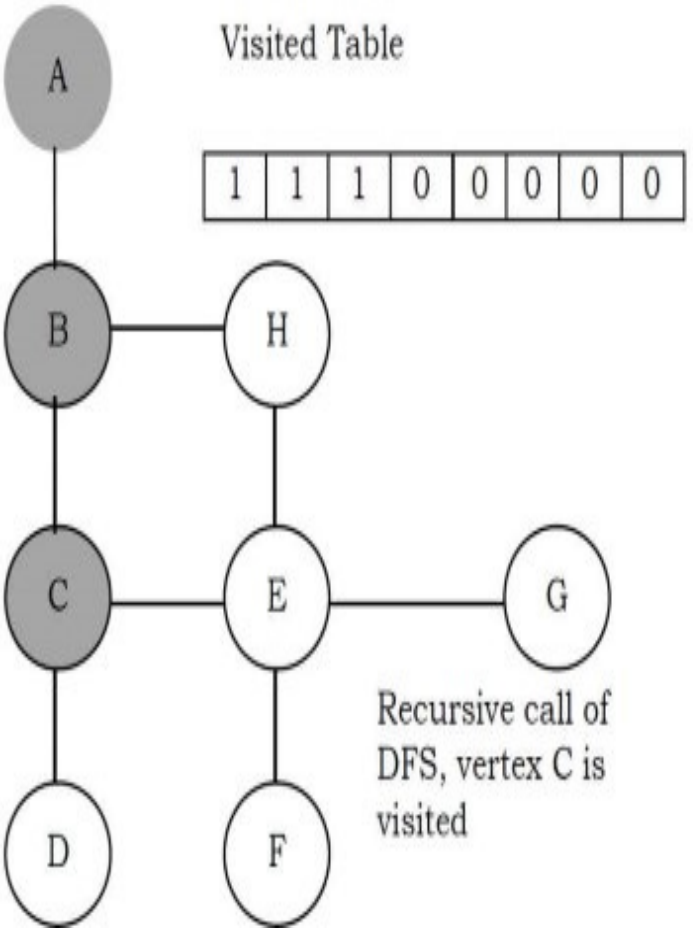
As an example, consider the following graph. We can see that sometimes an edge leads to an

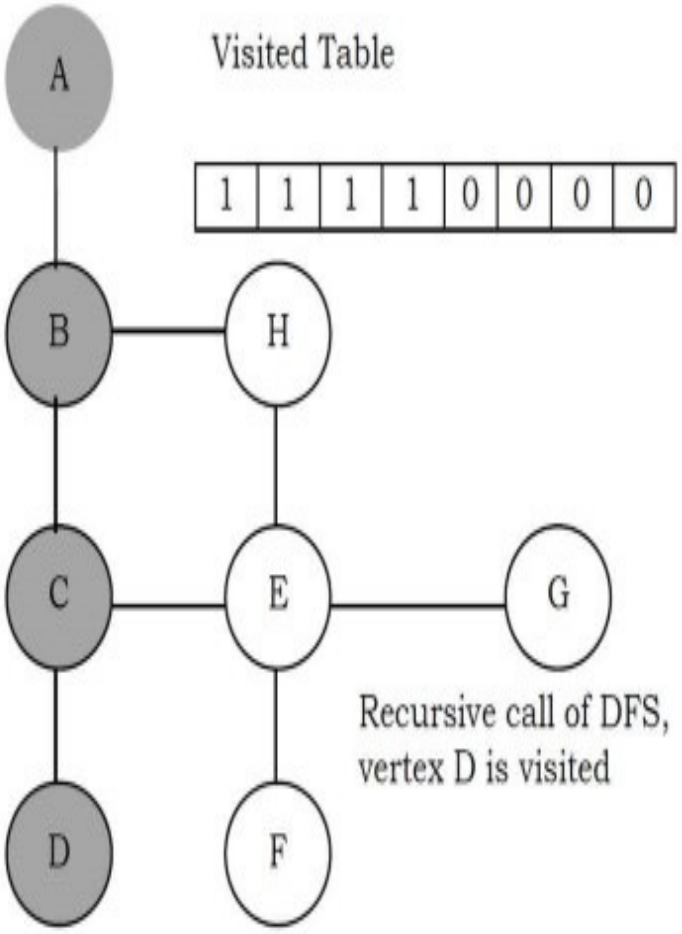
already discovered vertex. These edges are called *back edges*, and the other edges are called *tree edges* because deleting the back edges from the graph generates a tree.

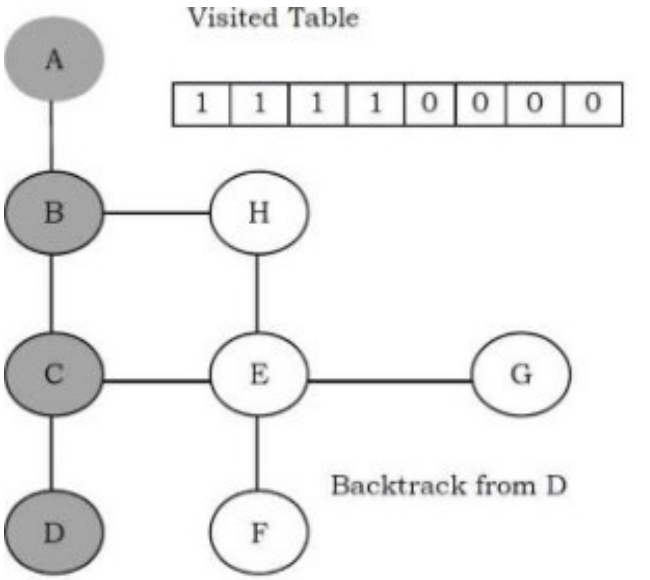
The final generated tree is called the DFS tree and the order in which the vertices are processed is called *DFS numbers* of the vertices. In the graph below, the gray color indicates that the vertex is visited (there is no other significance). We need to see when the Visited table is updated.

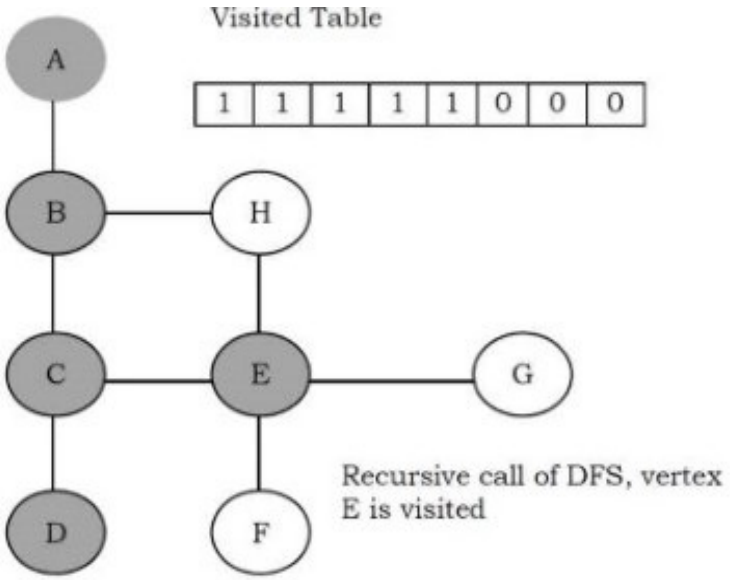

Visited Table
1 0 0 0 0 0 0 0
A
B
H
C
E
G
D
F
Starting vertex A is marked visited

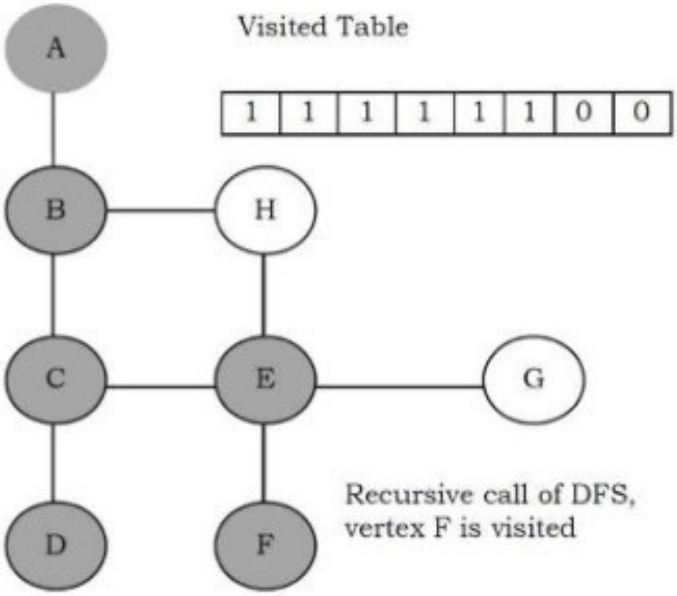

Visited Table
1 1 0 0 0 0 0 0
A
B
H
C
E
G
D
F
Vertex B is visited

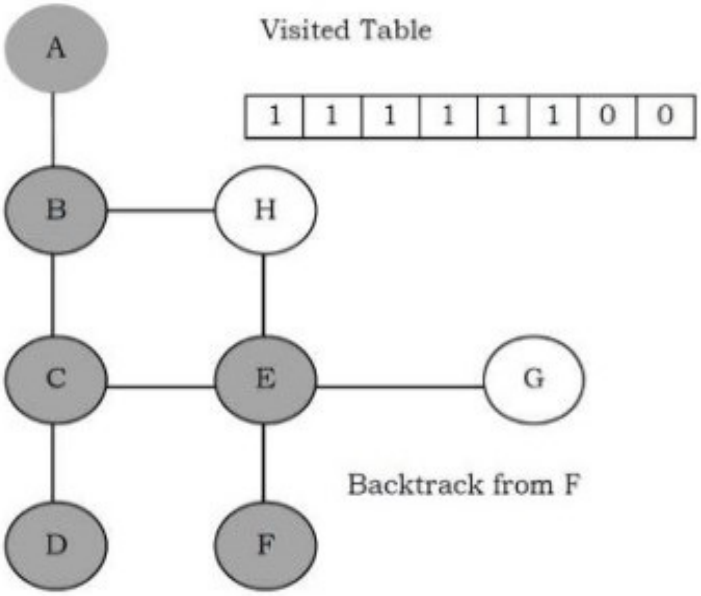

Visited Table
1 1 1 0 0 0 0 0
A
B
H
C
E
G
D
F
Recursive call of DFS, vertex C is visited

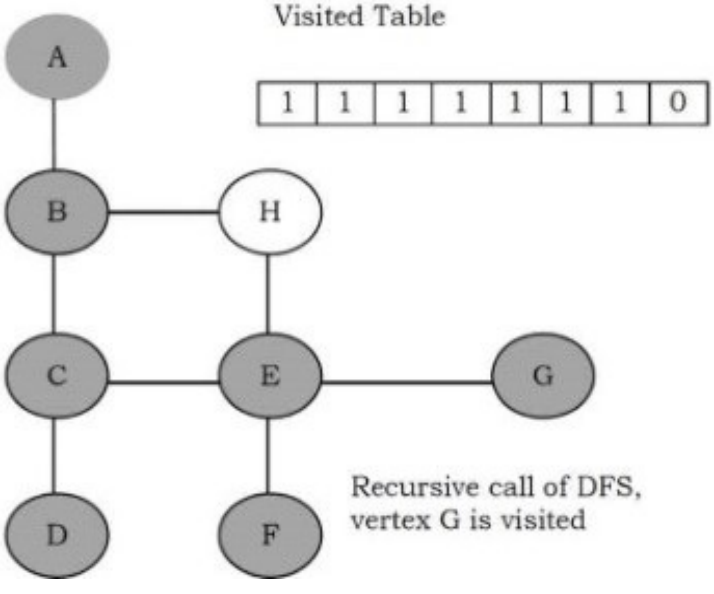

Visited Table
1 1 1 1 0 0 0 0
A
B
H
C
E
G
D
F
Recursive call of DFS, vertex D is visited

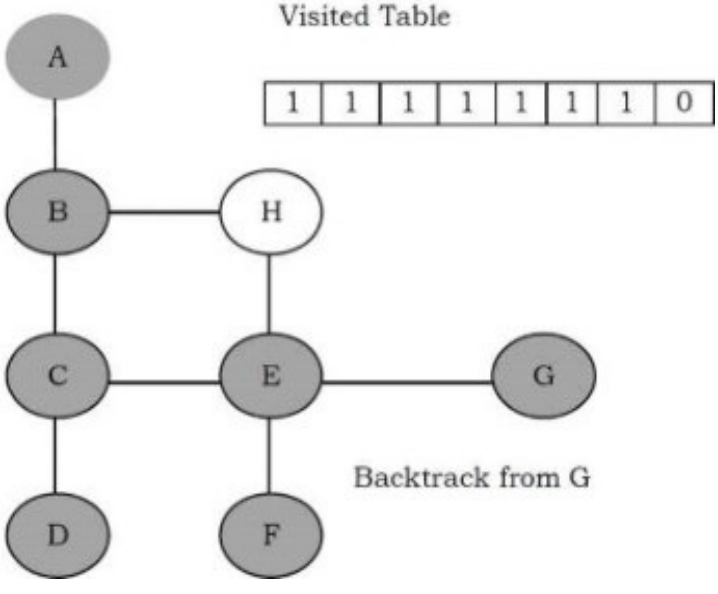

Visited Table
1 1 1 1 0 0 0 0
A
B
H
C
E
G
D
F
Backtrack from D

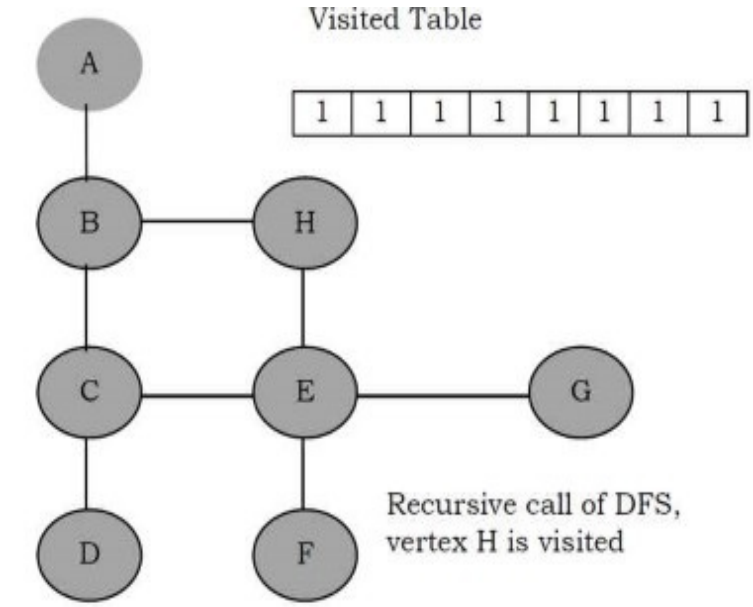

Visited Table
1 1 1 1 1 0 0 0
A
B
H
C
E
G
D
F
Recursive call of DFS, vertex E is visited

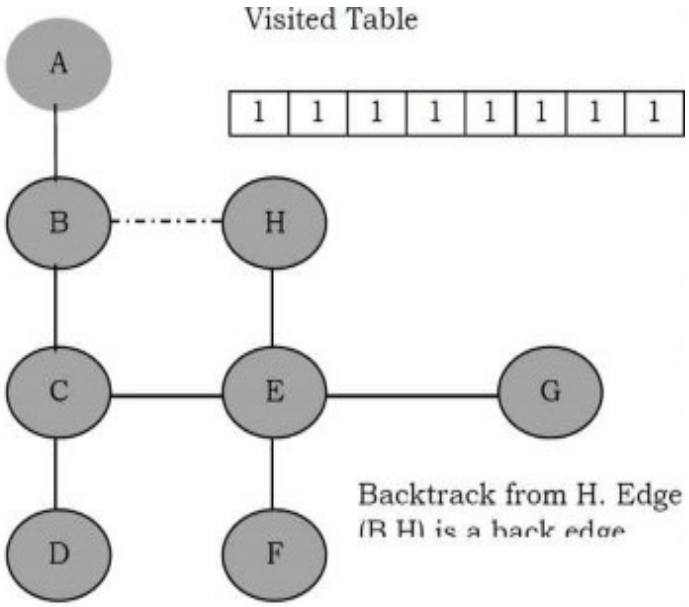

Visited Table
1 1 1 1 1 1 0 0
A
B
H
C
E
G
D
F
Recursive call of DFS, vertex F is visited

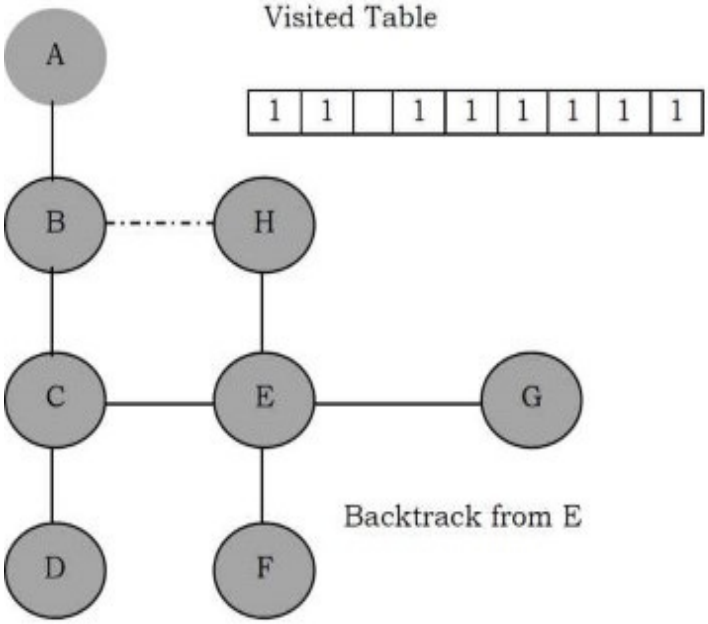

Visited Table
1 1 1 1 1 1 0 0
A
B
H
C
E
G
D
F
Backtrack from F

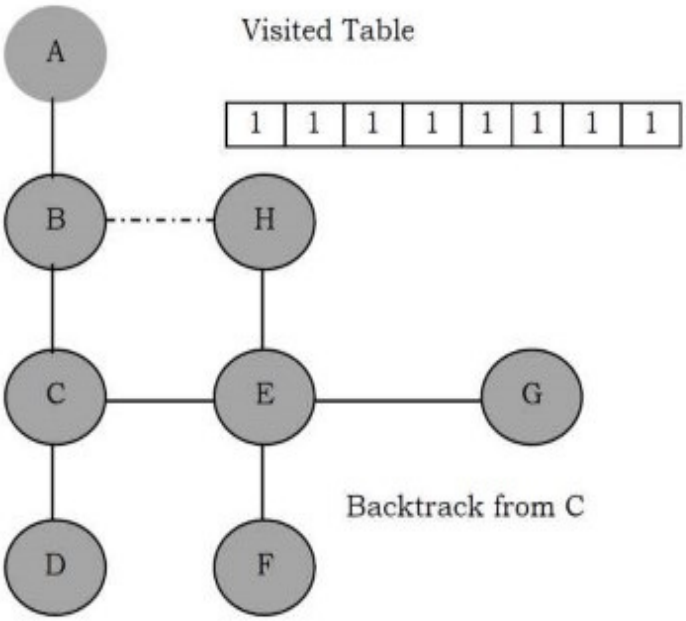

Visited Table
1 1 1 1 1 1 1 0
A
B
H
C
E
G
D
F
Recursive call of DFS, vertex G is visited

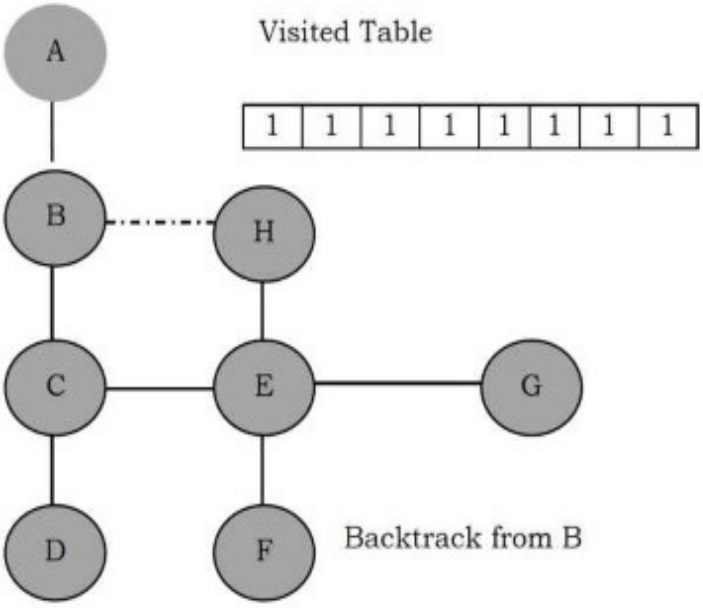

Visited Table
1 1 1 1 1 1 1 0
A
B
H
C
E
G
D
F
Backtrack from G

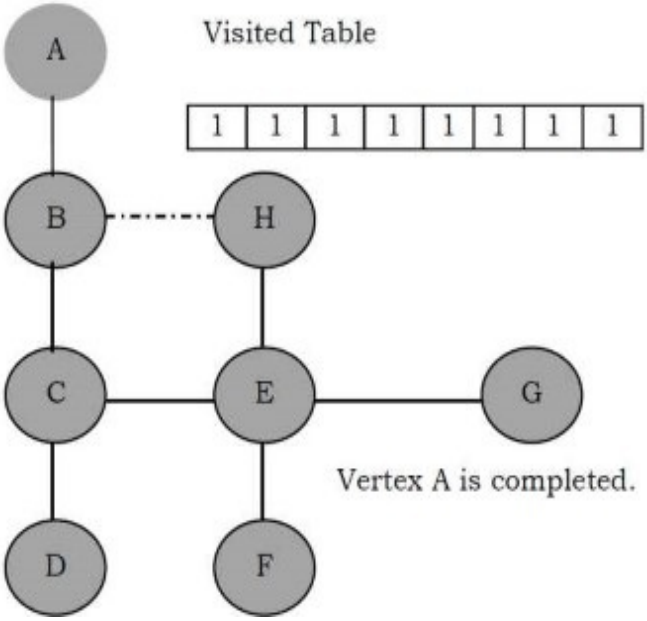

Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Recursive call of DFS,
vertex H is visited


Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Backtrack from H. Edge


Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Backtrack from E


Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Backtrack from C


Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Backtrack from B


Visited Table
1 1 1 1 1 1 1 1
A
B
H
C
E
G
D
F
Vertex A is completed.

From the above diagrams, it can be seen that the DFS traversal creates a tree (without back edges) and we call such tree a *DFS tree*. The above algorithm works even if the given graph has connected components.

The time complexity of DFS is O($V + E$), if we use adjacency lists for representing the graphs. This is because we are starting at a vertex and processing the adjacent nodes only if they are not visited. Similarly, if an adjacency matrix is used for a graph representation, then all edges adjacent to a vertex can't be found efficiently, and this gives O(V^2) complexity.

Applications of DFS

- Topological sorting
- Finding connected components
- Finding articulation points (cut vertices) of the graph
- Finding strongly connected components
- Solving puzzles such as mazes

For algorithms refer to *Problems Section.*

Breadth First Search [BFS]

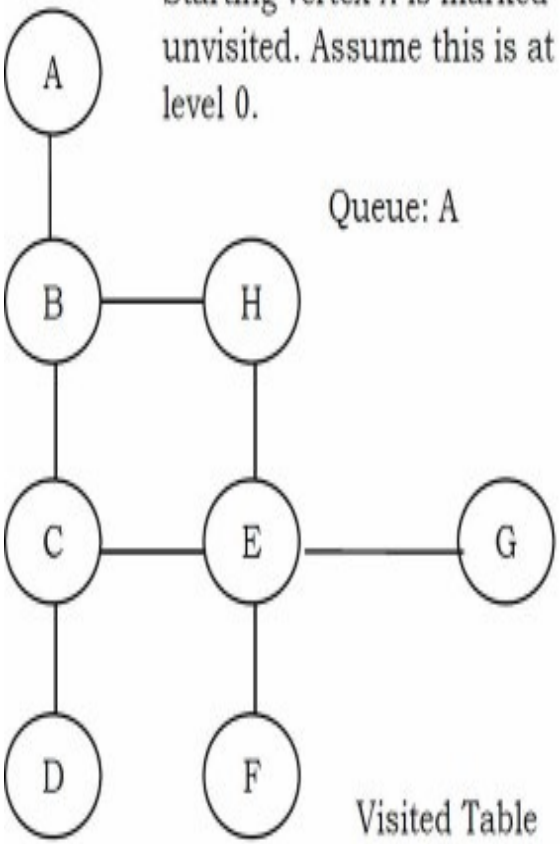
The BFS algorithm works similar to *level – order* traversal of the trees. Like *level – order* traversal, BFS also uses queues. In fact, *level – order* traversal got inspired from BFS. BFS works level by level. Initially, BFS starts at a given vertex, which is at level 0. In the first stage it visits all vertices at level 1 (that means, vertices whose distance is 1 from the start vertex of the graph). In the second stage, it visits all vertices at the second level. These new vertices are the ones which are adjacent to level 1 vertices.

BFS continues this process until all the levels of the graph are completed. Generally *queue* data structure is used for storing the vertices of a level.

As similar to DFS, assume that initially all vertices are marked *unvisited* (*false*). Vertices that have been processed and removed from the queue are marked *visited* (*true*). We use a queue to represent the visited set as it will keep the vertices in the order of when they were first visited. The implementation for the above discussion can be given as:

```
void BFS(struct Graph *G, int u) {
    int v;
    struct Queue *Q = CreateQueue();

    EnQueue(Q, u);

   while(!IsEmptyQueue(Q)) {
        u = DeQueue(Q);

        Process u; //For example, print

        Visited[s]=1;

        /* For example, if the adjacency matrix is used for representing the  graph,
        then the condition be used for finding unvisited adjacent vertex of u  is:
        if( !Visited[v] && G→Adj[u][v] ) */
        for each unvisited adjacent node v of u {
            EnQueue(Q, v);
        }
    }
}

void BFSTraversal(struct Graph *G) {
    for (int i = 0; i< G→V;i++)
        Visited[i]=0;

    //This loop is required if the graph has more than one component
   for (int i = 0; i< G→V;i++)
        if(!Visited[i])
            BFS(G. i):
}
```

As an example, let us consider the same graph as that of the DFS example. The BFS traversal can be shown as:

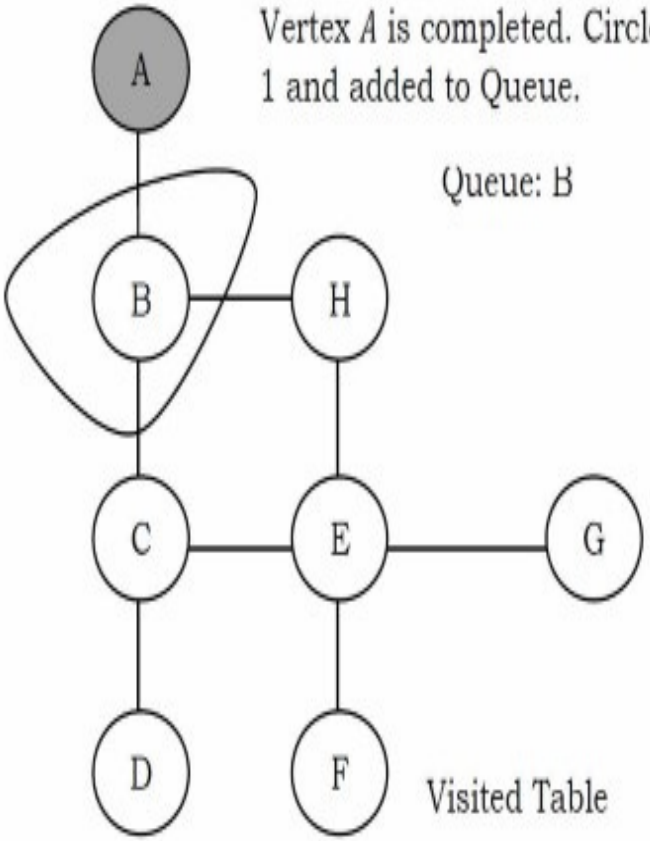


Starting vertex *A* is marked unvisited. Assume this is at level 0.

Queue: A

Visited Table

0	0	0	0	0	0	0



Vertex *A* is completed. Circled part is level 1 and added to Queue.

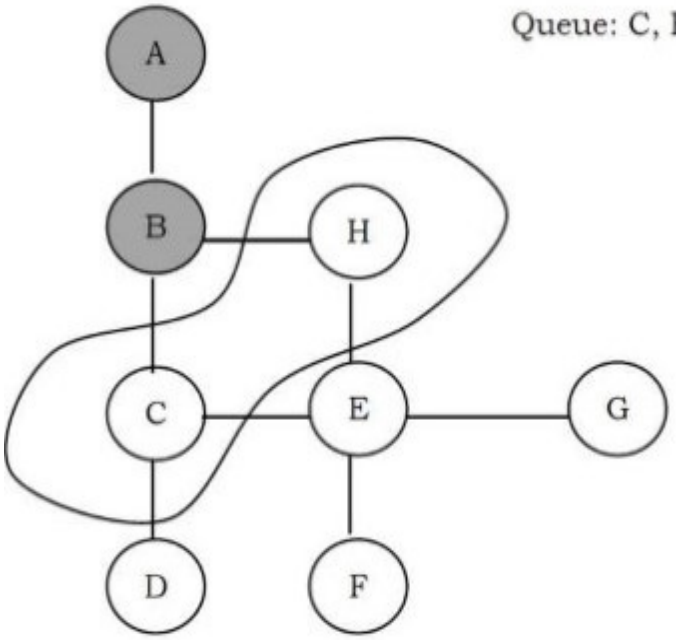
Queue: B

Visited Table

1	0	0	0	0	0	0

B is completed. Selected part is level 2 (add to Queue).

Queue: C, H

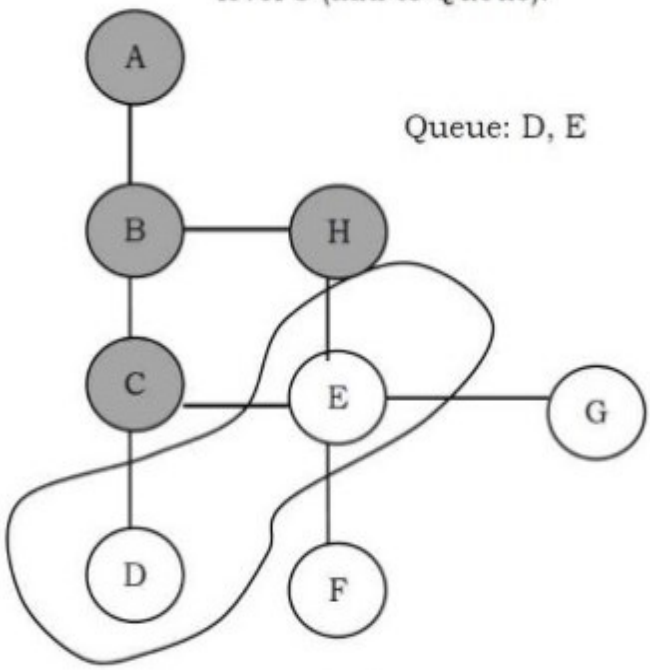


Visited Table

1	1	0	0	0	0	0

Vertices C and H are completed. Circled part is level 3 (add to Queue).

Queue: D, E

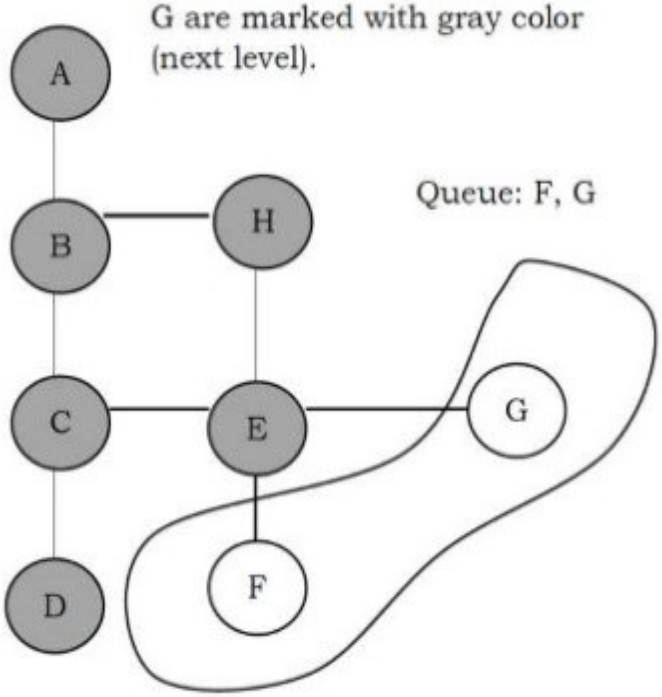


Visited Table

1	1	0	0	0	0	1

D and E are completed. F and G are marked with gray color (next level).

Queue: F, G

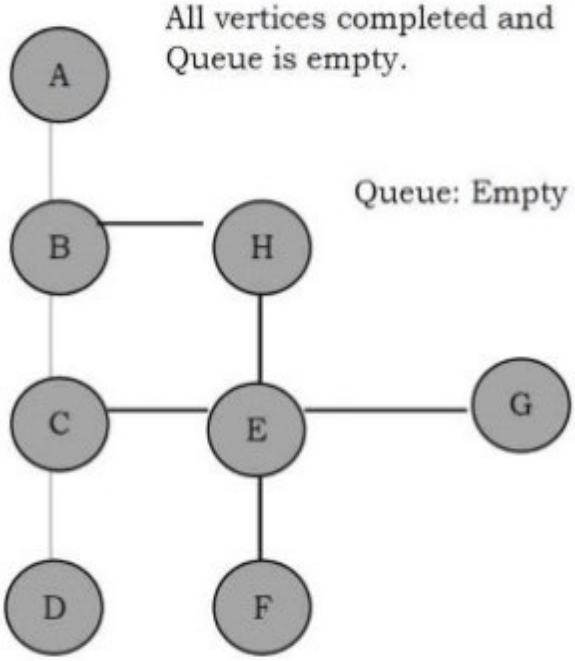


Visited Table

1	1	1	1	0	0	1

All vertices completed and Queue is empty.

Queue: Empty



Visited Table

1	1	1	1	1	1	1

Time complexity of BFS is O($V + E$), if we use adjacency lists for representing the graphs, and O(V^2) for adjacency matrix representation.

Applications of BFS

- Finding all connected components in a graph
- Finding all nodes within one connected component
- Finding the shortest path between two nodes
- Testing a graph for bipartiteness

Comparing DFS and BFS

Comparing BFS and DFS, the big advantage of DFS is that it has much lower memory requirements than BFS because it's not required to store all of the child pointers at each level. Depending on the data and what we are looking for, either DFS or BFS can be advantageous. For example, in a family tree if we are looking for someone who's still alive and if we assume that person would be at the bottom of the tree, then DFS is a better choice. BFS would take a very long time to reach that last level.

The DFS algorithm finds the goal faster. Now, if we were looking for a family member who died a very long time ago, then that person would be closer to the top of the tree. In this case, BFS finds faster than DFS. So, the advantages of either vary depending on the data and what we are looking for.

DFS is related to preorder traversal of a tree. Like *preorder* traversal, DFS visits each node before its children. The BFS algorithm works similar to *level – order* traversal of the trees.

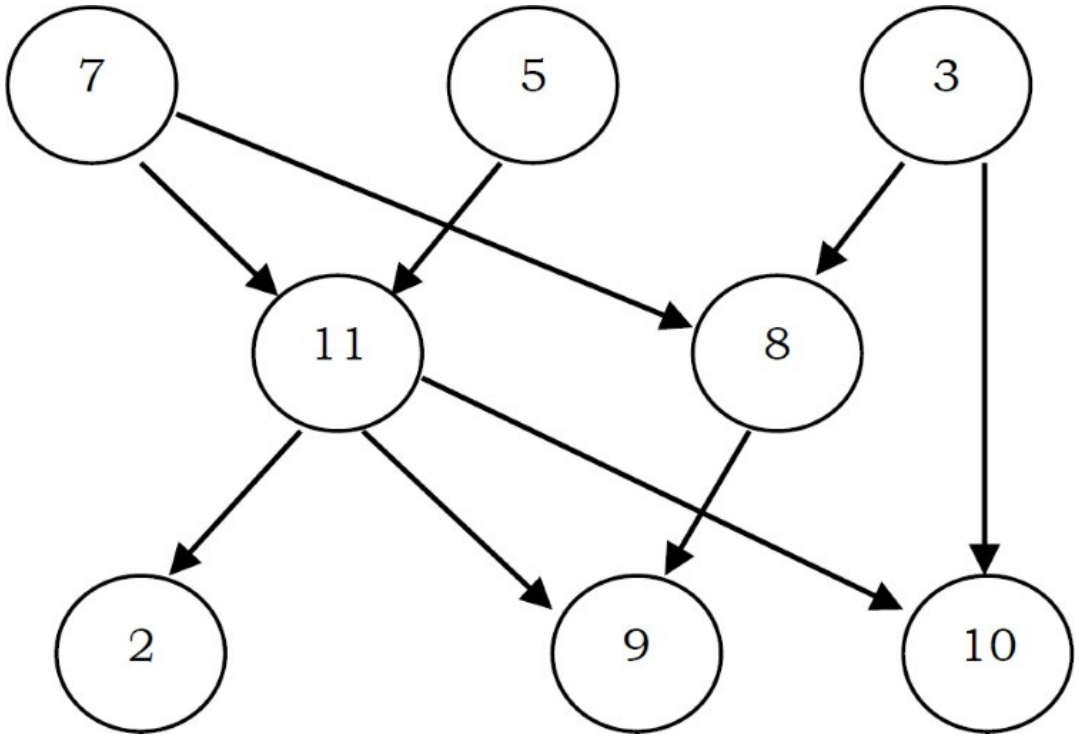
If someone asks whether DFS is better or BFS is better, the answer depends on the type of the problem that we are trying to solve. BFS visits each level one at a time, and if we know the solution we are searching for is at a low depth, then BFS is good. DFS is a better choice if the solution is at maximum depth. The below table shows the differences between DFS and BFS in terms of their applications.

Applications	DFS	BFS
Spanning forest, connected components, paths, cycles	Yes	Yes
Shortest paths		Yes
Minimal use of memory space	Yes	

9.6 Topological Sort

Topological sort is an ordering of vertices in a directed acyclic graph [DAG] in which each node comes before all nodes to which it has outgoing edges. As an example, consider the course prerequisite structure at universities. A directed *edge* (*v,w*) indicates that course *v* must be completed before course *w*. Topological ordering for this example is the sequence which does not violate the prerequisite requirement. Every DAG may have one or more topological orderings. Topological sort is not possible if the graph has a cycle, since for two vertices *v* and w on the cycle, *v* precedes w and w precedes *v*.

Topological sort has an interesting property. All pairs of consecutive vertices in the sorted order are connected by edges; then these edges form a directed Hamiltonian path [refer to *Problems Section*] in the DAG. If a Hamiltonian path exists, the topological sort order is unique. If a topological sort does not form a Hamiltonian path, DAG can have two or more topological orderings. In the graph below: 7, 5, 3, 11, 8, 2, 9, 10 and 3, 5, 7, 8, 11, 2, 9, 10 are both topological orderings.



Initially, *indegree* is computed for all vertices, starting with the vertices which are having indegree 0. That means consider the vertices which do not have any prerequisite. To keep track of vertices with indegree zero we can use a queue.

All vertices of indegree 0 are placed on queue. While the queue is not empty, a vertex *v* is removed, and all edges adjacent to *v* have their indegrees decremented. A vertex is put on the queue as soon as its indegree falls to 0. The topological ordering is the order in which the vertices DeQueue.

The time complexity of this algorithm is $O(|E| + |V|)$ if adjacency lists are used.